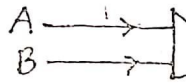
Digital Logic Design

Basic Logic Operation: There are three Basic Logic operation. They are: AND, OR, NOT

SHUVO
CSE-27

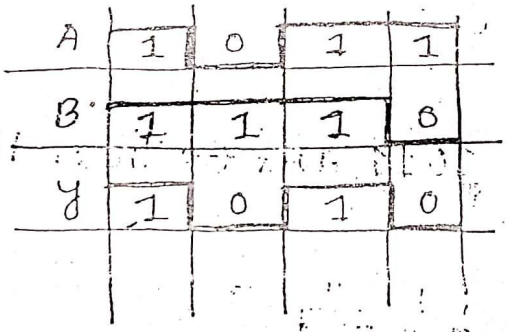
Logical Expression of AND: $y = AB$



Truth Table:

A	B	y
0	0	0
0	1	0
1	0	0
1	1	1

Timing Diagram: If A=1011 and B= 1110 then Find out the timing diagram of y



∴ y = 1010

Logical Expression of OR: $y = A + B$

Truth Table:

A	B	y
0	0	0
0	1	1
1	0	1
1	1	1

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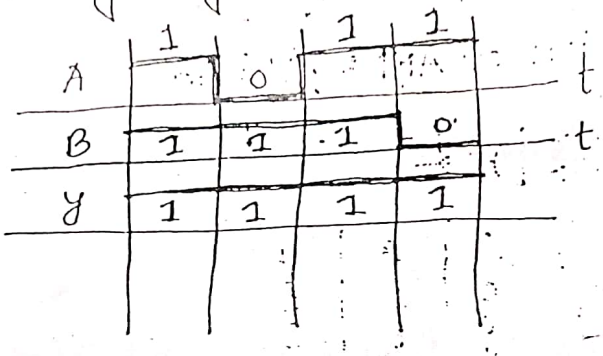
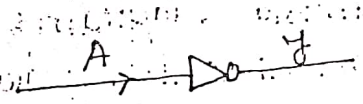
Timing Diagram: If A=1011 and B=1110 then find out the timing diagram of y.



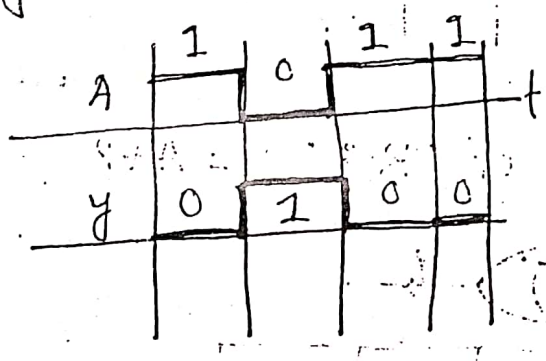
Logical Expression of NOT: $y = \bar{A}(A')$

Truth Table:

A	y
0	1
1	0



Timing Diagram: If A=1011 and B= then find out timing diagram of y.



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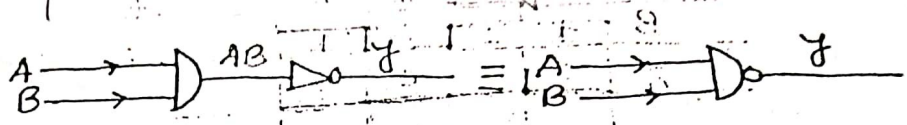
Additional Logic operations: There are two Additional Logic operation they are NAND and NOR gate.

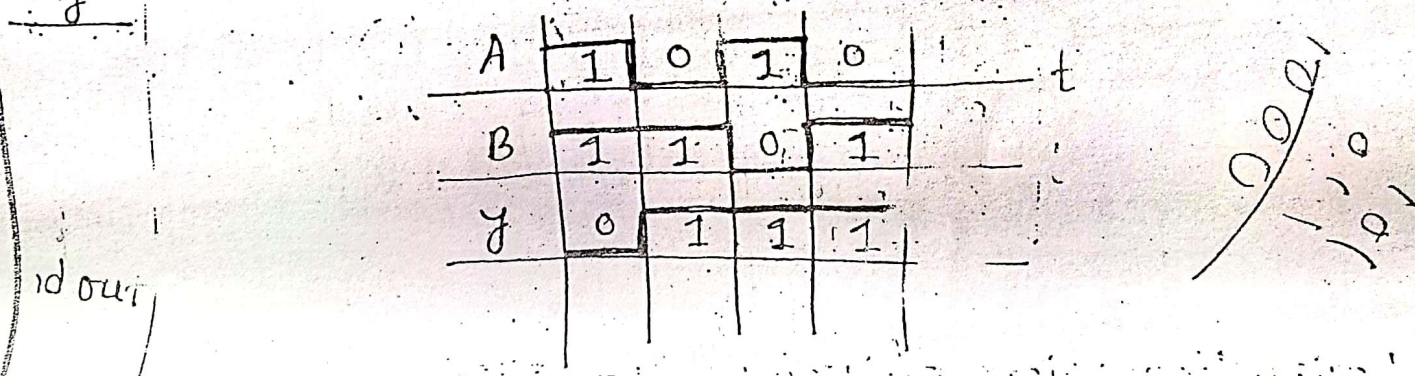
Logical Expression of NAND: $y = \overline{AB}$



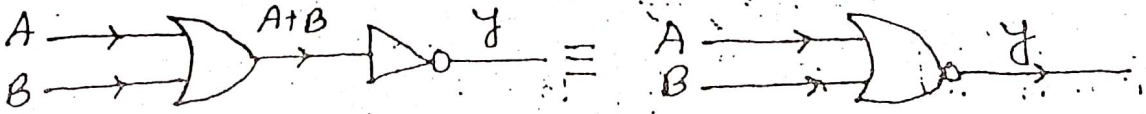
Truth Table:

A	B	y
0	0	1
0	1	1
1	0	1
1	1	0

Timing Diagram: If A = 1010 and B = 1101 then find the timing diagram of y.



Logical Expression of NOR: $y = \overline{A+B}$



Truth Table:

A	B	y
0	0	1
0	1	0
1	0	0
1	1	0

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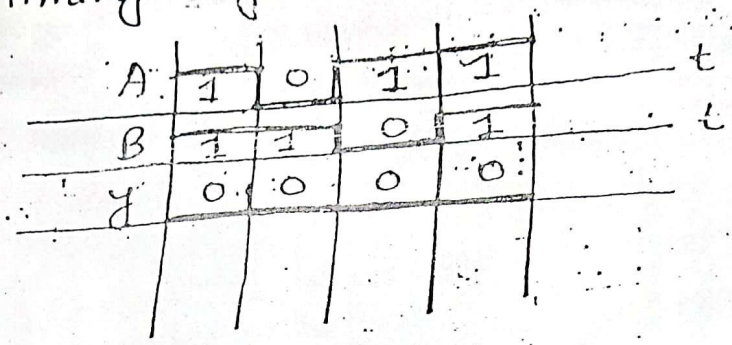
Timing Diagram: If A = 1011 and B = 1101 then find out the timing diagram of Y

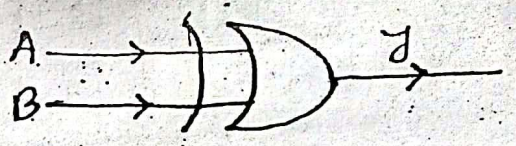


Exclusive Logic operation: There two Exclusive Logic operation. They are XOR, NOR.

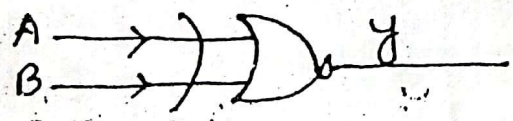
Logical Expression of XOR: $Y = A \oplus B$

Truth Table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



Logical Expression of XNOR: $Y = \overline{A \oplus B}$



Truth Table:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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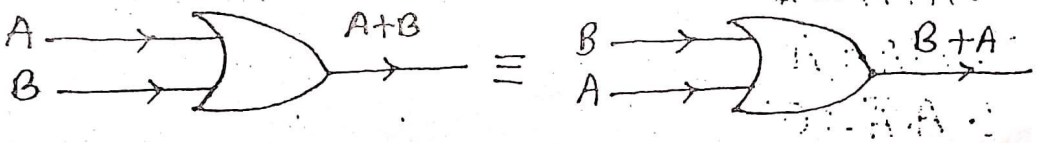
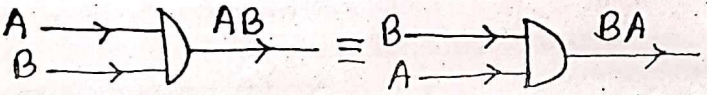
...ean Algebra : There are ... of boolean Algebra
... given laws.
(i) Commutative Law.
(ii) Associative Law.
(iii) Distributive Law.

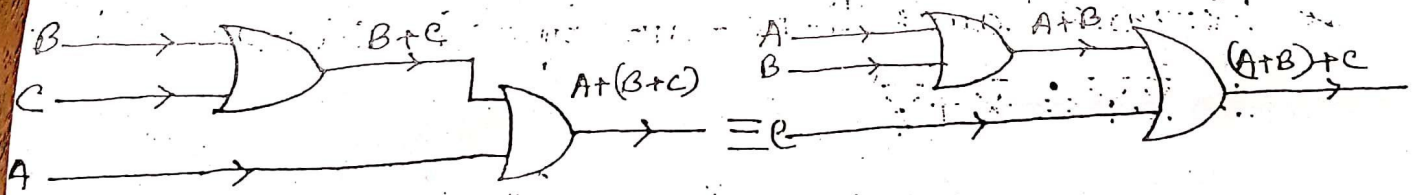
Commutative law : (i) $A+B = B+A$



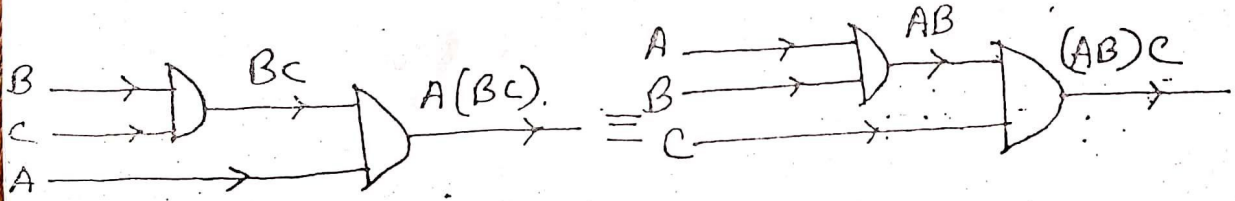
(ii) $AB = BA$



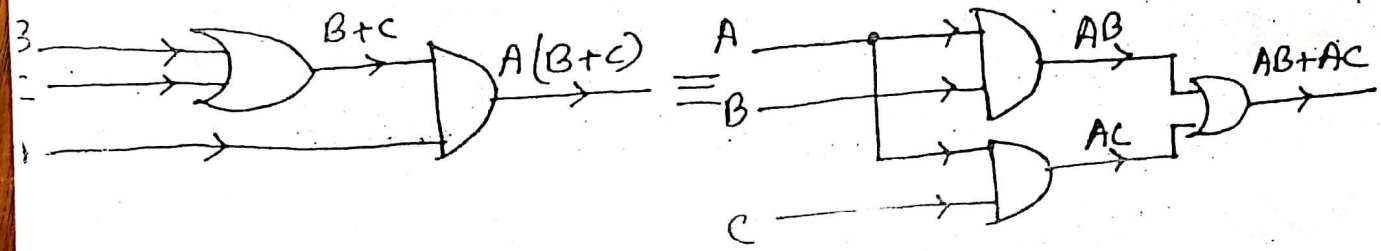
Associative law : (i) $A+(B+C) = (A+B)+C$



(ii) $A(BC) = (AB)C$



Distributive law : (i) $A(B+C) = AB+AC$



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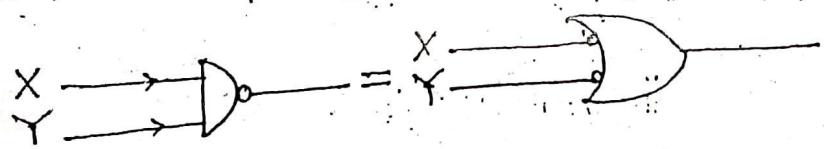
Boolean Rules: There are some Boolean rules. They are given below.

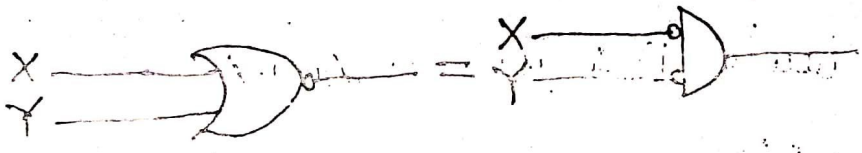
1. $A+0=A$
2. $A+1=1$
3. $A\cdot 0=0$
4. $A\cdot 1=A$
5. $A+A=A$
6. $A+\bar{A}=1$
7. $A\cdot A=A$
8. $A\cdot\bar{A}=0$
9. $A+AB=A$
10. $A+\bar{A}B=A+B$
11. $\bar{\bar{A}}=A$
12. $(A+B)(A+C)=A+BC$ ✓
13. $(\bar{A}\bar{B}+AB)=\overline{A\oplus B}$.
14. $(\bar{A}B+A\bar{B})=A\oplus B$.

De Morgan's Law: There are two De Morgan's Law.

1: $\overline{X\cdot Y}=\bar{X}+\bar{Y}$



2. $\overline{X+Y}=\bar{X}\cdot\bar{Y}$



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de Morgan's Theorem :

$$\overline{(A+B)\cdot\overline{C}} + \overline{AB+\overline{D}} = \overline{(A+B)}\cdot\overline{\overline{C}}\cdot\left(\overline{AB+\overline{D}}\right)$$

$$= \left(\overline{(A+B)}+\overline{C}\right)\cdot(AB+\overline{D})$$

$$= (A+B+C)(AB+\overline{D})$$

$$\overline{A+B\overline{C}} + D\left(\overline{E+F}\right) = \left(\overline{A+B\overline{C}}\right)\cdot D\left(\overline{E+F}\right)$$

$$= (A+B\overline{C})\cdot D\cdot\left(\overline{E+F}\right)$$

$$= (A+B\overline{C})\left(\overline{D}+\left(\overline{\overline{E+F}}\right)\right)$$

$$= (A+B\overline{C})\left(\overline{D}+E+F\right)$$

Simplification of Boolean Expression :

(i) $AB + A(B+C) + B(B+C)$

$= AB + AB + AC + BB + BC$

$= AB + AC + B + B\overline{C}$

$= AB + AC + B(1+C)$

$= AB + AC + B$

$= B(A+1) + AC$

$= B + AC$.

(ii) $\left[A\overline{B}(C+BD) + \overline{A}\overline{B}\right]C$

$= \left[A\overline{B}C + A\overline{B}BD + \overline{A}\overline{B}\right]C$

$= A\overline{B}CC + A\overline{B}BDC + \overline{A}\overline{B}C$

$= A\overline{B}C + \overline{A}\overline{B}C$

$= (A+\overline{A})\overline{B}C$

$= \overline{B}C$.

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Findout the truth table $y = A + BC$.

A	B	C	BC	A+BC
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Findout the truth table $y = AB\bar{C} + A\bar{B}C + \bar{A}BC$

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

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ndard form of Boolean expression: There are two
em of Boolean expression. They are —
(i) Sum of product (SOP)
(ii) Product of sum (POS)

nvert the following SOP into SSOP : [SSOP = Standard Sum of product —

$Y = AB + ABC + \bar{B}C$

$= AB(C + \bar{C}) + ABC + (A + \bar{A})\bar{B}C$

$= ABC + AB\bar{C} + ABC + A\bar{B}C + \bar{A}\bar{B}C$

$= ABC + AB\bar{C} + A\bar{B}C + \bar{A}\bar{B}C$ *

Convert the following POS into SPOS :

$Y = (A+B)(\bar{B}+C)(A+B+C)$

$= (A+B+C\bar{C})(A\bar{A}+\bar{B}+C)(A+B+C)$

$= (A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+\bar{B}+C)$ [Boolean rule ::

* SOP এর ক্ষেত্রে $A = 1, B = 1$ । $A = 0, B = 0$
* POS এর ক্ষেত্রে $A = 0, B = 0$ । $\bar{A} = 1, \bar{B} = 1$

vert the following SOP into SPOS :

$= AB + \bar{B}C + \bar{A}\bar{B}\bar{C}$

$= ABC + AB\bar{C} + A\bar{B}C + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}$

$= \underset{7}{ABC} + \underset{6}{AB\bar{C}} + \underset{5}{A\bar{B}C} + \underset{1}{\bar{A}\bar{B}C} + \underset{0}{\bar{A}\bar{B}\bar{C}}$

$= \underset{2}{(A+\bar{B}+C)}\ \underset{3}{(A+\bar{B}+\bar{C})}\ \underset{4}{(\bar{A}+B+C)}$ [SPOS Convention

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Convert the ssop into spos :

$y = \bar{A}\bar{B}\bar{C} + AB\bar{C} + \bar{A}B\bar{C} + A\bar{B}C$ ⊗ [0, 3, 4, 7]

$\bar{A}\bar{B}\bar{C}$ = 001 = 1 ; $AB\bar{C}$ = 110 = 6 ; $\bar{A}B\bar{C}$ = 010 = 2 ; $A\bar{B}C$ = 101 = 5

$= (A+B+C)$ 000 = 0 ; $(A+\bar{B}+\bar{C})$ 011 = 3 ; $(\bar{A}+B+C)$ 100 = 4 ; $(\bar{A}+\bar{B}+\bar{C})$ 111 = 7

Karnaugh Map : 3-variable karnaugh map -----

$y = A\bar{B} + B\bar{C} + AB\bar{C} + A\bar{B}\bar{C}$

$= A\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} + A\bar{B}\bar{C} + \bar{A}B\bar{C} + AB\bar{C} + A\bar{B}\bar{C}$

$= A\bar{B}C + A\bar{B}\bar{C} + AB\bar{C} + \bar{A}B\bar{C}$ [∵ Any one will grante in the condition]

AB \ C	0	1
00	0	
01	1	0
11	1	→ Group 1.
10	1	1 — Group 2.

∴ $y = B\bar{C} + A\bar{B}$

$\bar{A}\bar{B}(C+\bar{C})$

	$\bar{B}$	B
$\bar{A}$		
A		

$C(B+\bar{B})$

$B\bar{C} + \bar{B}C$

BC \ C	C
00	C
01	C
10	C
11	C

(05)

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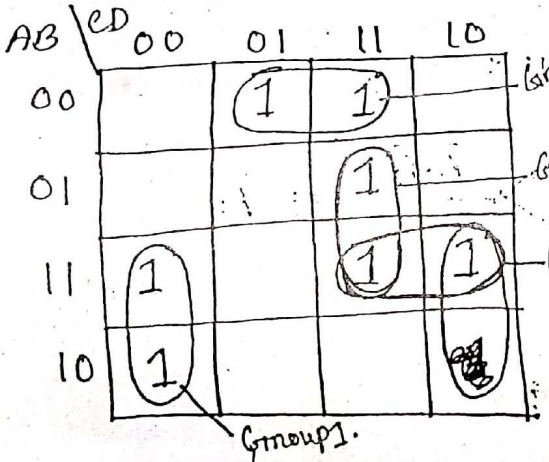
augh Map : 4-variable karnaugh map ----

$y = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}BCD + AB\bar{C}\bar{D} + AB\bar{C}\bar{D} + ABC\bar{D} + ABCD$

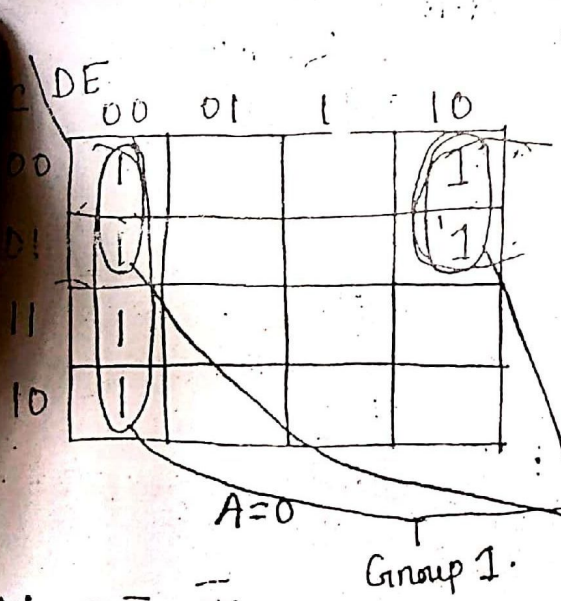


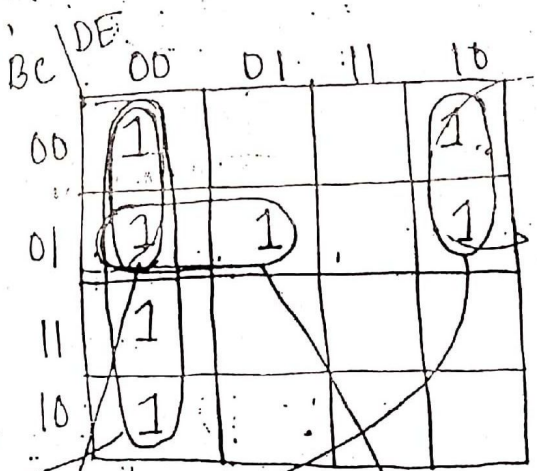
[If any one remain alone, it will join with any adjacent one]

$\therefore y = A\bar{C}\bar{D} + \bar{A}\bar{B}D + BCD + ABC$

karnaugh map : 5-variable karnaugh map ----

$y = \bar{A}BC\bar{D}\bar{E} + A\bar{B}C\bar{D}E + \bar{A}B\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}C\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}\bar{D}\bar{E} + \bar{A}\bar{B}\bar{C}D\bar{E} + \bar{A}\bar{B}CD\bar{E} + A\bar{B}\bar{C}\bar{D}\bar{E} + A\bar{B}C\bar{D}\bar{E} + AB\bar{C}\bar{D}\bar{E} + A\bar{B}\bar{C}D\bar{E} + A\bar{B}CD\bar{E}$





$\therefore y = \bar{D}\bar{E} + \bar{B}\bar{D}\bar{E} + A\bar{B}C\bar{D}$

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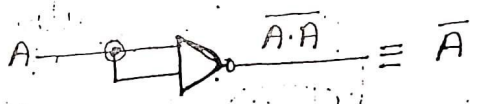
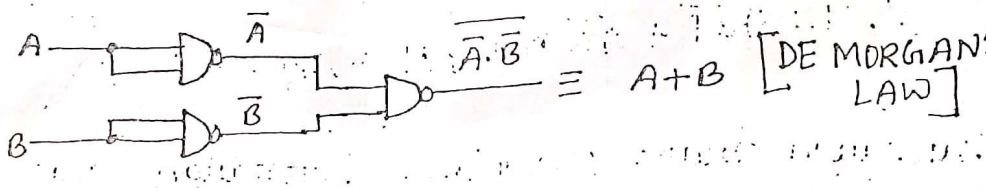
Universal gats: There are two universal gats. They are NAND gats and NOR gats.

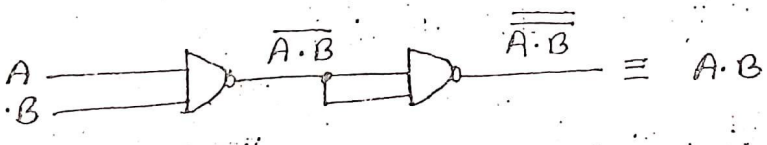
(i) From NAND to NOT:



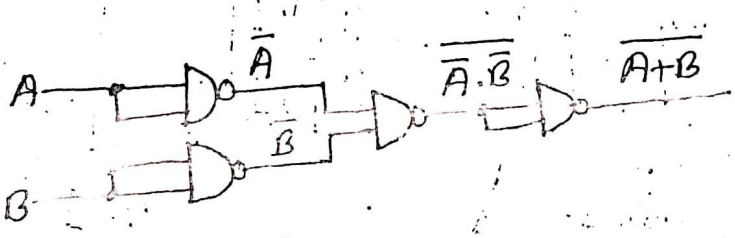
(ii) From NAND to OR:



(iii) From NAND to AND:



(iv) From NAND to NOR:



* As From NAND gats we can prove any kinds of gats, NAND gats is called Universal gats.

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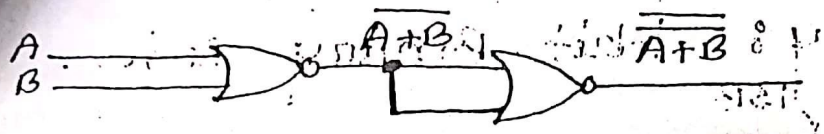
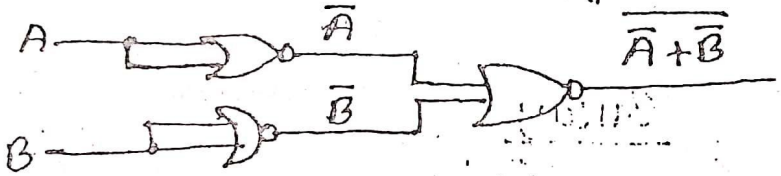
om NOR to AND:

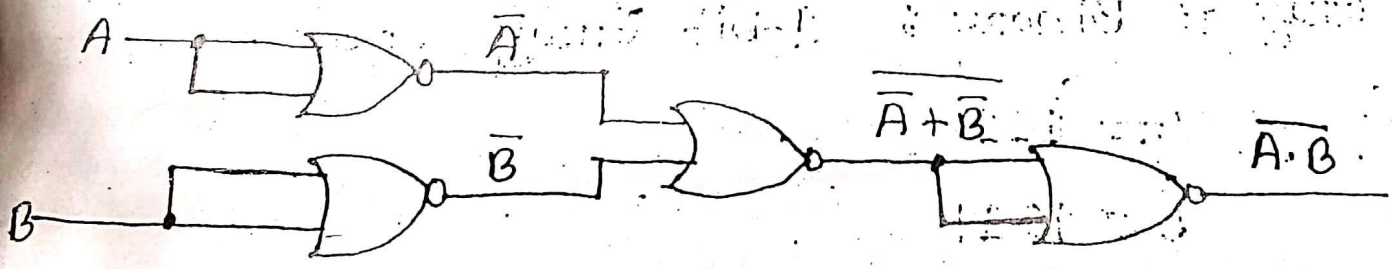


(ii) From NOR to AND:



(iii) From NOR to NOT:

(iv) From NOR to NAND:



e can express any gate from NOR gate. So NOR
is an Universal gate.

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Convertion of Binary to Gray code and vice

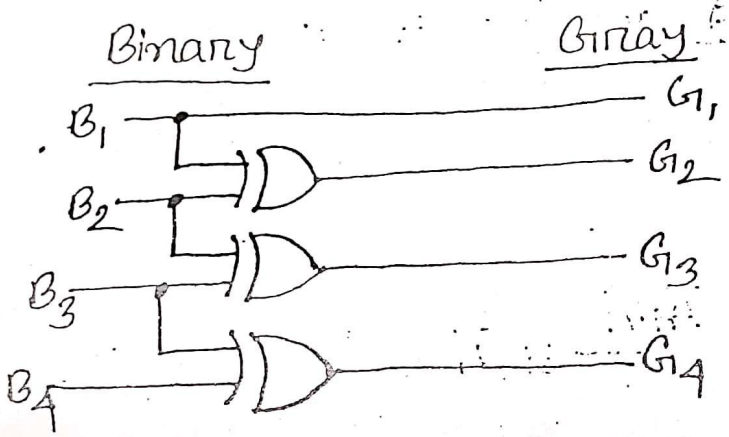
(i) Binary to Gray : 4-bit Binary code.

MSB LSB

B = 1011

G = 1110

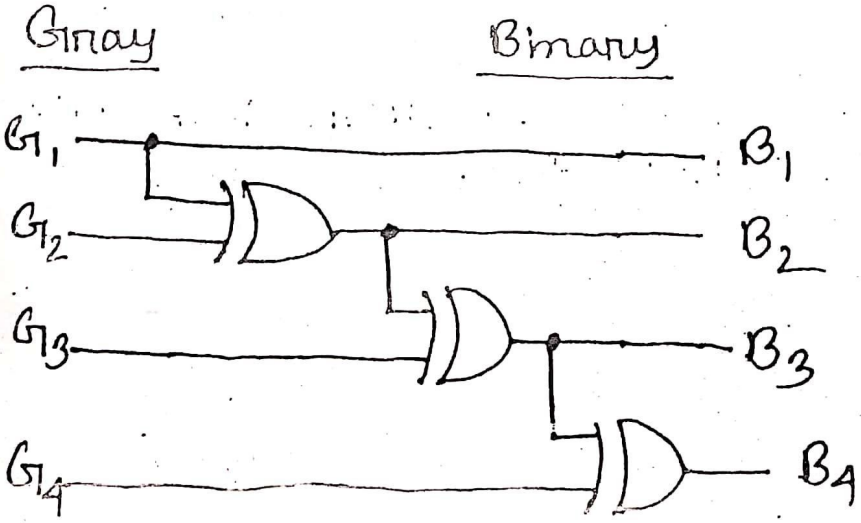
[It acts like exclusi



(ii) Gray to Binary : 4-bit Gray code.

G = 1110

B = 1011



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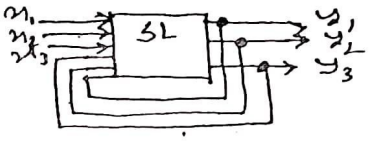
...ion of combinational logic : There are two types of functional combination logic. They are - - - -

(i) Combinational logic. (CL)

(ii) Sequential logic (SL)



Adder : There are two adder they are —

(i) Half Adder.

(ii) Full Adder.

(i) Half adder :

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Expression :

$\therefore S = \bar{A}B + A\bar{B}$

$= A \oplus B$

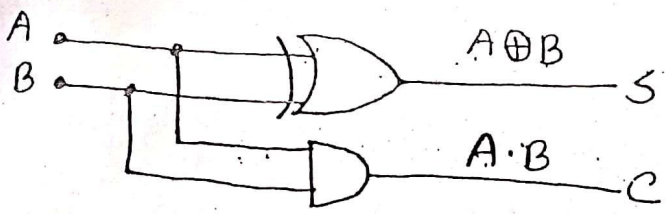
$\therefore C = A \cdot B$

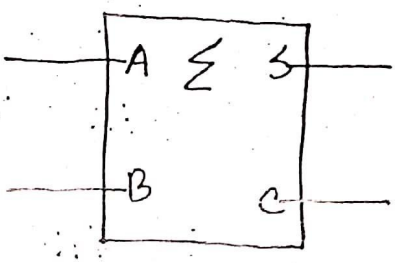
[$S = 0+0 = 0$ Cout = 0

$S = 0+1 = 1$ Cout = 0

$S = 1+1 = 1$ Cout = 1]

Block diagram :





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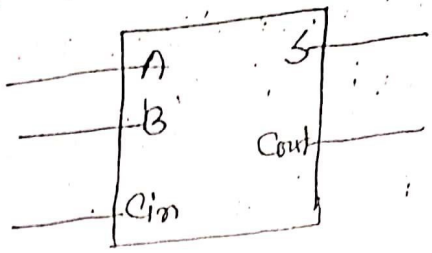
(ii) Full adder:



A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

[S বের করতে হলে At First Binary সংখ্যা গুলো যোগ করতে হবে because then যোগফলকে Binary তে convert করতে হবে]

[S = 1 + 1 + 1 = 3
B = 11
∴ S = 1, C = 1]

pression:

$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$

$= (\bar{A}B + A\bar{B})\bar{C}_{in} + (\bar{A}\bar{B} + AB)C_{in}$

$= (A \oplus B)\bar{C}_{in} + \overline{(A \oplus B)}C_{in}$ [$\because \bar{A}\bar{B} + AB = \overline{A \oplus B}$]

$= (A \oplus B) \oplus C_{in}$

$= A \oplus B \oplus C_{in}$

$C_{out} = A\bar{B}C_{in} + A\bar{B}C_{in} + \bar{A}BC_{in} + ABC_{in}$

$= AB(\bar{C}_{in} + C_{in}) + (\bar{A}B + A\bar{B})C_{in}$

$= AB + (A \oplus B)C_{in}$

11

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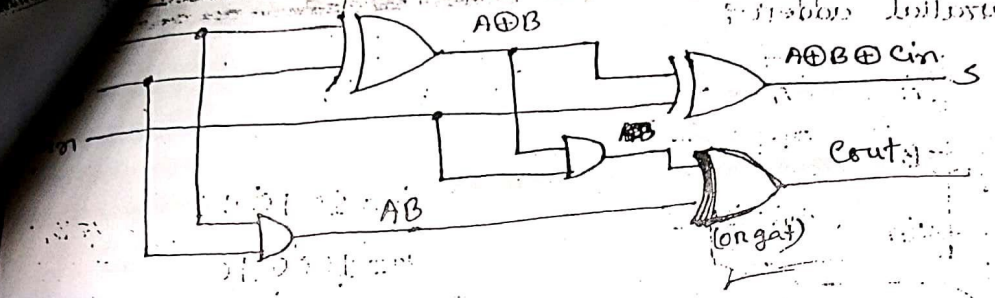
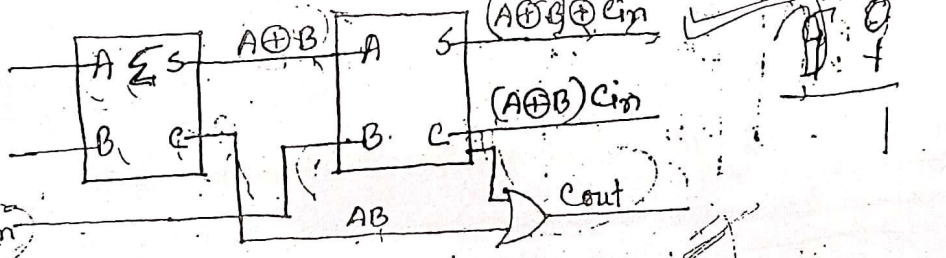
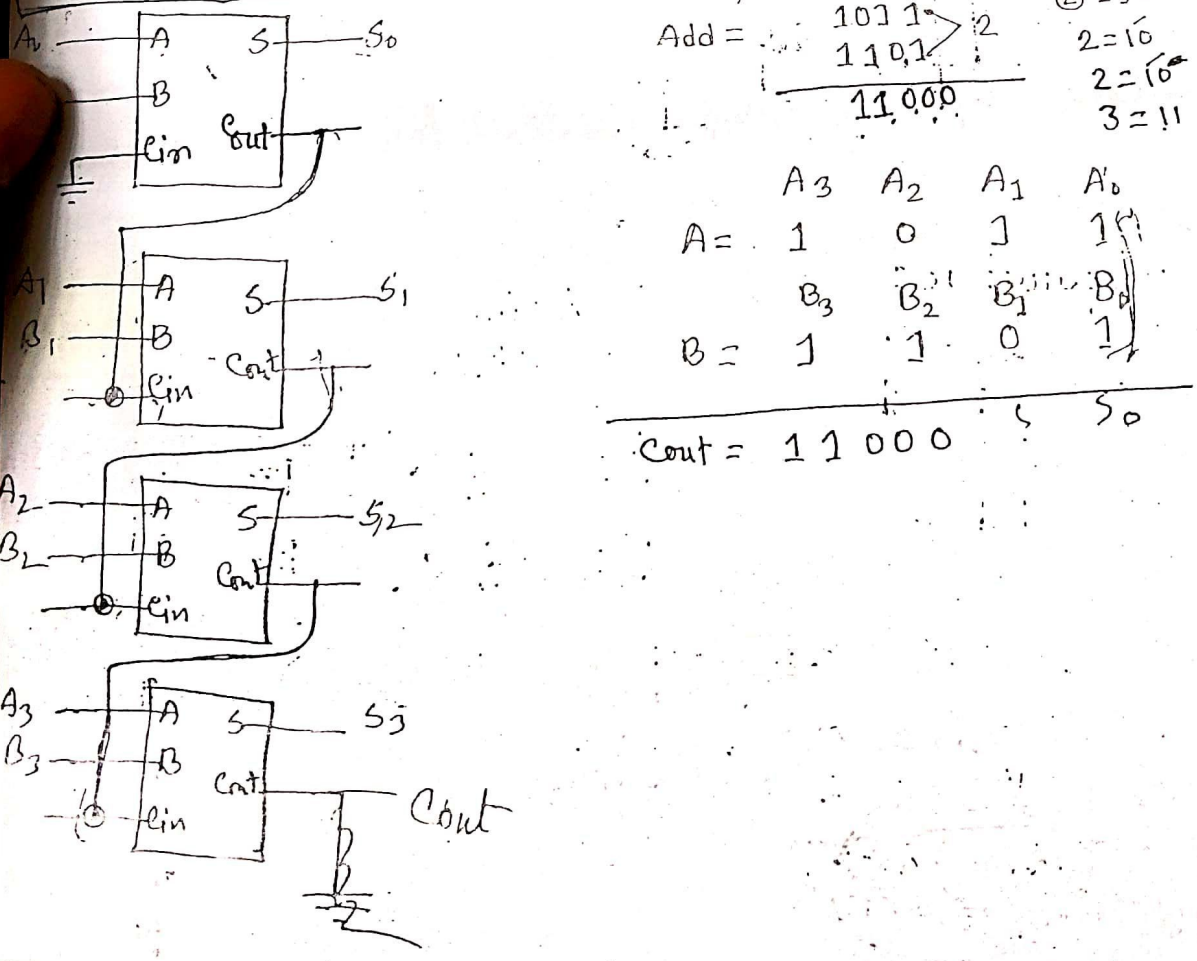


Block diagram:



Draw the 4-bit parallal adder:



Add =
1011 → 2
1101
11000

② = 10
2 = 10
2 = 10
3 = 11

	A_3	A_2	A_1	A_0
A =	1	0	1	1
	B_3	B_2	B_1	B_0
B =	1	1	0	1
Cout =	1	1	0	0	0
				S_1	S_0

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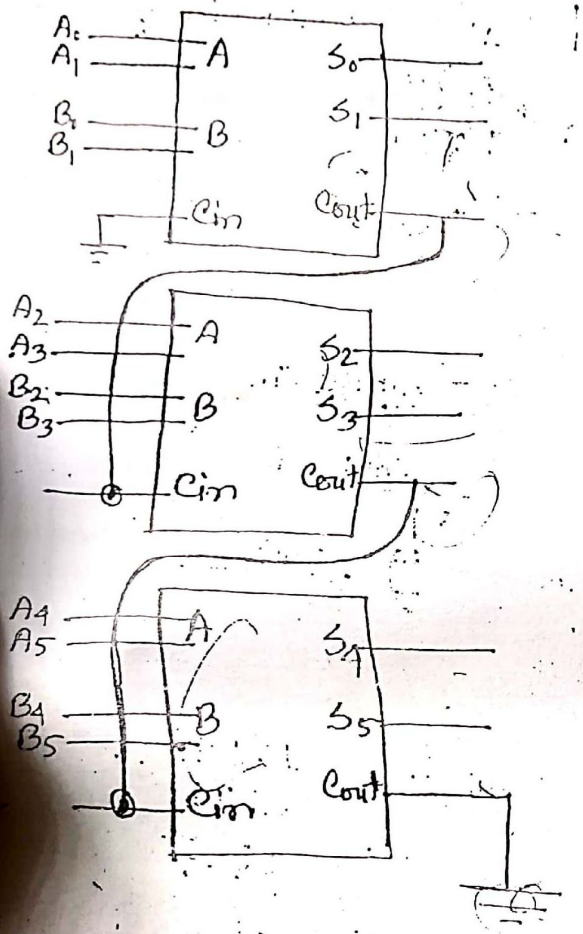
Draw the block diagram of a 6-bit parallal adder using 2 bit parallal adder:



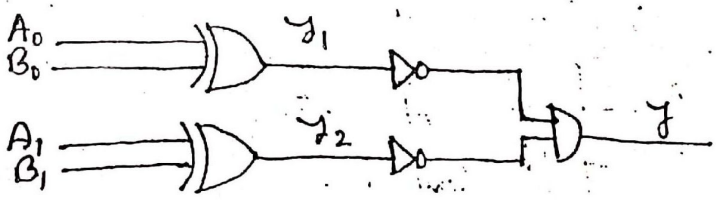
A = 101011
B = 100010
1 001101

Comparator:

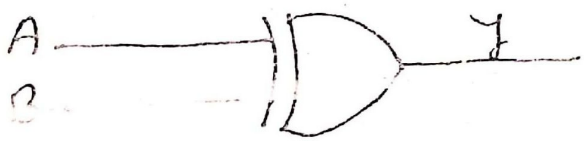
A	B	
0	0	= 0
0	1	= 1
1	0	= 1
1	1	= 0

$A = A_1 A_0$

$B = B_1 B_0$



If $y = 0$ then $A = B$



else $A \neq B$.

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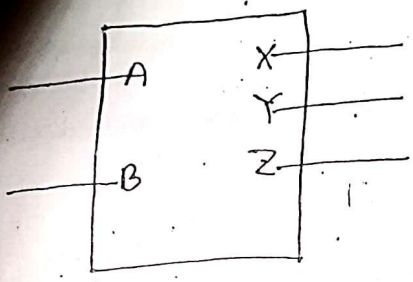
Comparator:



Truth table:

A	B	X	Y	Z
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

This truth table follows following rules:

(i) If $A = B$ then $X = 1$, $Y = 0$, $Z = 0$

(ii) If $A > B$ then $Y = 1$, $X = 0$, $Z = 0$

(iii) If $A < B$ then $Z = 1$, $X = 0$, $Y = 0$

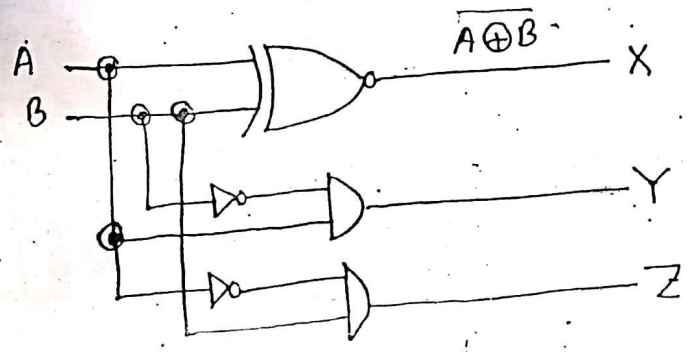
Expression:

$$X = \bar{A}\bar{B} + AB = \overline{A \oplus B}$$

$$Y = \bar{A}B$$

$$Z = A\bar{B}$$

Circuit Diagram:



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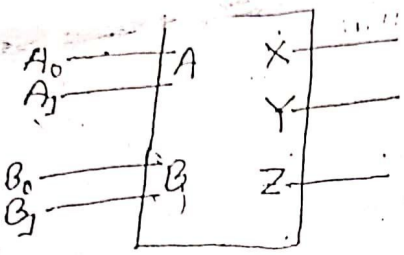
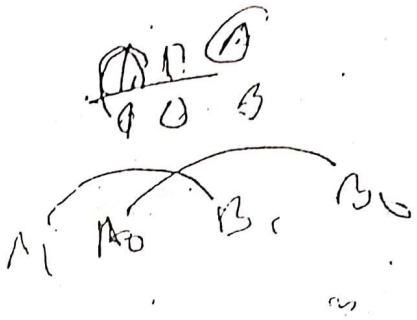
2-bit Comparator :





Truth table :

A_0	A_1	B_0	B_1	X	Y	Z	
0	0	0	0	1	0	0	[00 = 0 00 = 0 A
0	0	0	1	0	0	1	[00 = 0, 01 = 1 A < B
0	0	1	0	0	0	1	[00 = 0, 10 = 2, A < B
0	0	1	1	0	0	1	[00 = 0, 11 = 3 A < B
0	1	0	0	0	1	0	[01 = 1, 00 = 0 A > B]
0	1	0	1	1	0	0	[01 = 1, 01 = 1, A = B]
0	1	1	0	0	0	1	[01 = 1, 10 = 2, A < B]
0	1	1	1	0	0	1	[01 = 1, 11 = 3, A < B]
1	0	0	0	0	1	0	[10 = 2, 00 = 0, A > B]
1	0	0	1	0	1	0	[10 = 2, 01 = 1, A > B]
1	0	1	0	1	0	0	[10 = 2, 10 = 2, A = B]
1	0	1	1	0	0	1	[10 = 2, 11 = 3, A < B]
1	1	0	0	0	1	0	[11 = 3, 00 = 0, A > B]
1	1	0	1	0	1	0	[11 = 3, 01 = 1, A > B]
1	1	1	0	0	1	0	[11 = 3, 10 = 2, A > B]
1	1	1	1	1	0	0	[11 = 3, 11 = 3, A = B]

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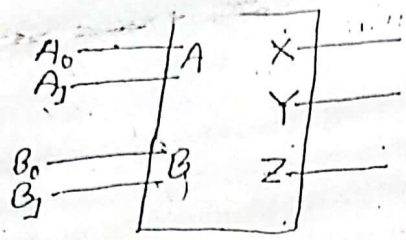
2-bit Comparator :



Truth table :

A_0	A_1	B_0	B_1	X	Y	Z
0	0	0	0	1	0	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	0	1	0
0	1	0	1	1	0	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	1	0
1	0	0	1	0	1	0
1	0	1	0	1	0	0
1	0	1	1	0	0	1
1	1	0	0	0	1	0
1	1	0	1	0	1	0
1	1	1	0	0	1	0
1	1	1	1	1	0	0

[00 = 0 00 = 0 A

[00 = 0, 01 = 1 A <

[00 = 0, 10 = 2, A <

[00 = 0, 11 = 3 A <

[01 = 1, 00 = 0 A > B]

[01 = 1, 01 = 1, A = B]

[01 = 1, 10 = 2, A < B]

[01 = 1, 11 = 3, A < B]

[10 = 2, 00 = 0, A > B]

[10 = 2, 01 = 1, A > B]

[10 = 2, 10 = 2, A = B]

[10 = 2, 11 = 3, A < B]

[11 = 3, 00 = 0, A > B]

[11 = 3, 01 = 1, A > B]

[11 = 3, 10 = 2, A > B]

[11 = 3, 11 = 3, A = B]

2-bit Comparator এর compare করার জন্য বাইনারী- মানের ডিজিগুলো মানের compare করতে হয়।

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Expression:

$$X = \bar{A}_0 \bar{A}_1 \bar{B}_0 \bar{B}_1 + \bar{A}_0 A_1 \bar{B}_0 B_1 + A_0 \bar{A}_1 B_0 \bar{B}_1 + A_0 A_1 B_0 B_1$$

$$= \bar{A}_0 \bar{B}_0 (\bar{A}_1 \bar{B}_1 + A_1 B_1) + A_0 B_0 (\bar{A}_1 \bar{B}_1 + A_1 B_1)$$

$$= (\bar{A}_1 \bar{B}_1 + A_1 B_1)(\bar{A}_0 \bar{B}_0 + A_0 B_0)$$

$$= \overline{(A_0 \oplus B_0)} \cdot \overline{(A_1 \oplus B_1)} \quad [\because \bar{A}\bar{B} + AB = \overline{A \oplus B}]$$

$$Y = \bar{A}_0 A_1 \bar{B}_0 \bar{B}_1 + \bar{A}_0 A_1 \bar{B}_0 \bar{B}_1 + A_0 \bar{A}_1 \bar{B}_0 B_1 + A_0 A_1 \bar{B}_0 \bar{B}_1 + A_0 \bar{A}_1 \bar{B}_0 B_1 + A_0 A_1 B_0 \bar{B}_1$$

$$= A_1 \bar{B}_1 (\bar{A}_0 \bar{B}_0 + A_0 B_0) + A_0 \bar{B}_0 (\bar{A}_1 \bar{B}_1 + A_1 B_1) + A_0 \bar{A}_1 \bar{B}_0 B_1 + A_0 A_1 \bar{B}_0 \bar{B}_1$$

$$= A_1 \bar{B}_1 \overline{(A_0 \oplus B_0)} + A_0 \bar{B}_0 \overline{(A_1 \oplus B_1)} + A_0 \bar{B}_0 (\bar{A}_1 B_1 + A \bar{B}_1)$$

$$= A_1 \bar{B}_1 \overline{(A_0 \oplus B_0)} + A_0 \bar{B}_0 \overline{(A_1 \oplus B_1)} + A_0 \bar{B}_0 (A_1 \oplus B_1)$$

$$= A_1 \bar{B}_1 \overline{(A_0 \oplus B_0)} + A_0 \bar{B}_0 \{\overline{(A_1 \oplus B_1)} + (A_1 \oplus B_1)\}$$

$$= A_1 \bar{B}_1 \overline{(A_0 \oplus B_0)} + A_0 \bar{B}_0$$

$$\bar{A}_0 \bar{A}_1 \bar{B}_0 B_1 + \bar{A}_0 \bar{A}_1 B_0 \bar{B}_1 + \bar{A}_0 A_1 B_0 B_1 + \bar{A}_0 A_1 B_0 \bar{B}_1 + \bar{A}_0 A_1 B_0 \bar{B}_1 + A_0 \bar{A}_1 B_0 B$$

$$= \bar{A}_1 B_1 (\bar{A}_0 \bar{B}_0 + A_0 B_0) + \bar{A}_0 B_0 (\bar{A}_1 \bar{B}_1 + A_1 B_1) + \bar{A}_0 B_0 (\bar{A}_1 B_1 + A_1 \bar{B}_1)$$

$$= \bar{A}_1 B_1 \overline{(A_0 \oplus B_0)} + \bar{A}_0 B_0 \overline{(A_1 \oplus B_1)} + \bar{A}_0 B (A_1 \oplus B_1)$$

$$= \bar{A}_1 B_1 \overline{(A_0 \oplus B_0)} + \bar{A}_0 B_0 \{\overline{(A_1 \oplus B_1)} + (A_1 \oplus B_1)\}$$

$$= \bar{A}_1 B_1 \overline{(A_0 \oplus B_0)} + \bar{A}_0 B_0$$

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2-bit comparator: We can write expression with Truth table.

$A = A_0 A_1$

$B = B_0 B_1$

(i) If $A_0 = B_0$ AND $A_1 = B_1$ then $A = B$.

(ii) If $A_0 > B_0$ OR $A_0 = B_0$ AND $A_1 > B_1$ then $A > B$

(iii) If $A_0 < B_0$ OR $A_0 = B_0$ AND $A_0 < B_1$ then $A < B$.

A_0	A_1	B_0	B_1	X	Y	Z

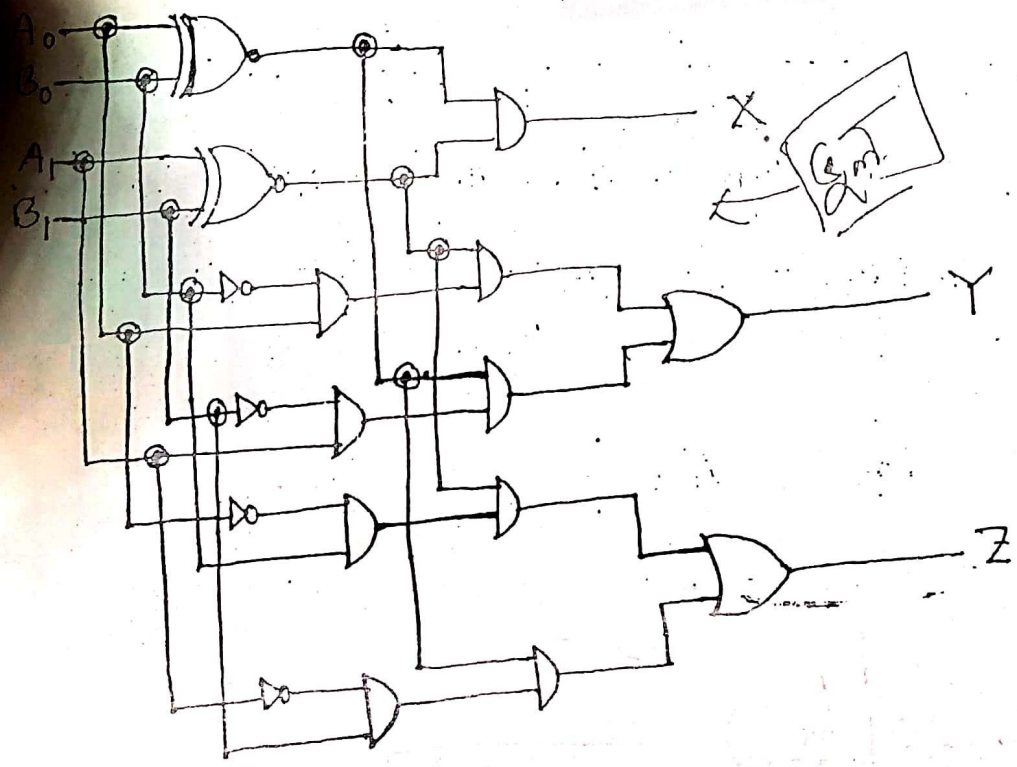
Expression:

$X = (\overline{A_0 \oplus B_0})\,(\overline{A_1 \oplus B_1})$

$Y = A_0 \overline{B_0}$ [illegible] $+ A_1 \overline{B_1}\,(\overline{A_0 \oplus B_0})$.

$Z = \overline{A_0} B_0$ [illegible] $+ \overline{A_1} B_1\,(\overline{A_0 \oplus B_0})$.

Circuit Diagram:



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Comparator:

$A = A_2 A_1 A_0$

$B = B_2 B_1 B_0$

If $A_2 = B_2$ AND $A_1 = B_1$ AND $A_0 = B_0$ then $A = B$.

If $A_2 > B_2$ OR $(A_2 = B_2$ AND $A_1 > B_1)$ OR $(A_2 = B_2$ AND $A_1 = B_1$ AND $A_0 > B_0)$ then $A > B$.

If $A_2 < B_2$ OR $(A_2 = B_2$ AND $A_1 < B_1)$ OR $(A_2 = B_2$ AND $A_1 = B_1$ AND $A_0 < B_0)$ then $A < B$.

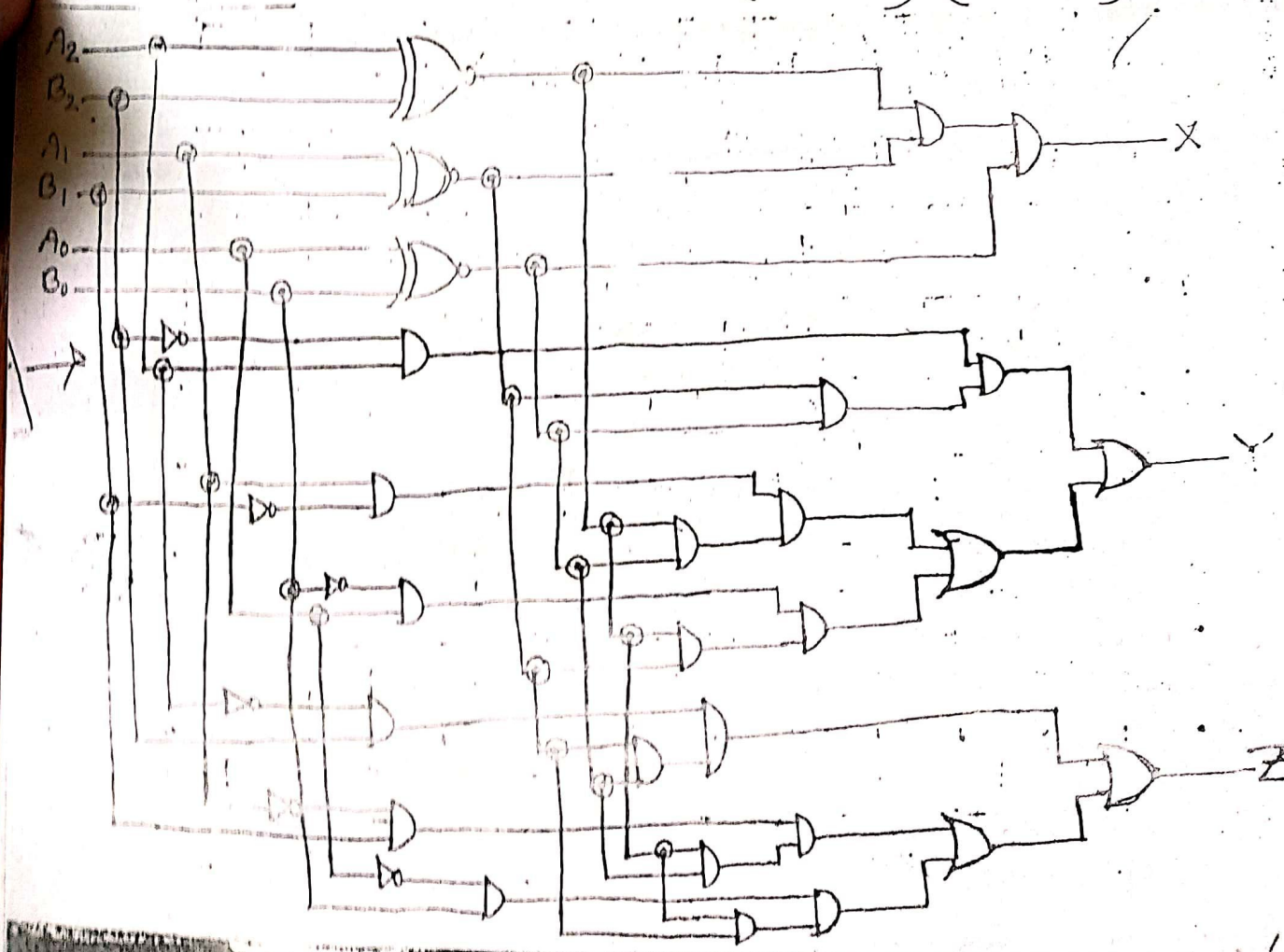
Expression:

$$X = (\overline{A_2 \oplus B_2}) \cdot (\overline{A_1 \oplus B_1}) \cdot (\overline{A_0 \oplus B_0})$$

$$Y = A_2 \overline{B_2} + A_1 \overline{B_1} (\overline{A_2 \oplus B_2}) + A_0 \overline{B_0} (\overline{A_2 \oplus B_2}) (\overline{A_1 \oplus \quad})$$

$$Z = \overline{A_2} B_2 + \overline{A_1} B_1 (\overline{A_2 \oplus B_2}) + \overline{A_0} B_0 (\overline{A_2 \oplus B_2}) (\overline{A_1 \oplus B_1})$$

Circuit Diagram:



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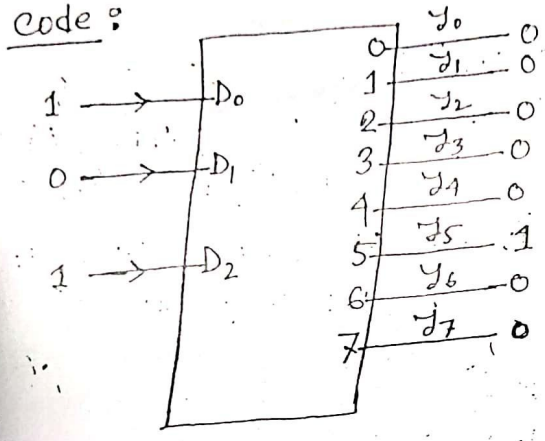
Decoder :

Encoder :

5 ⟷ 101

Code :



3-bit decoder :

Truth table :

D_0	D_1	D_2	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

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Expressions:

$Y_0 = \bar{D}_0 \bar{D}_1 \bar{D}_2$

$Y_1 = \bar{D}_0 \bar{D}_1 D_2$

$Y_2 = \bar{D}_0 D_1 \bar{D}_2$

$Y_3 = \bar{D}_0 D_1 D_2$

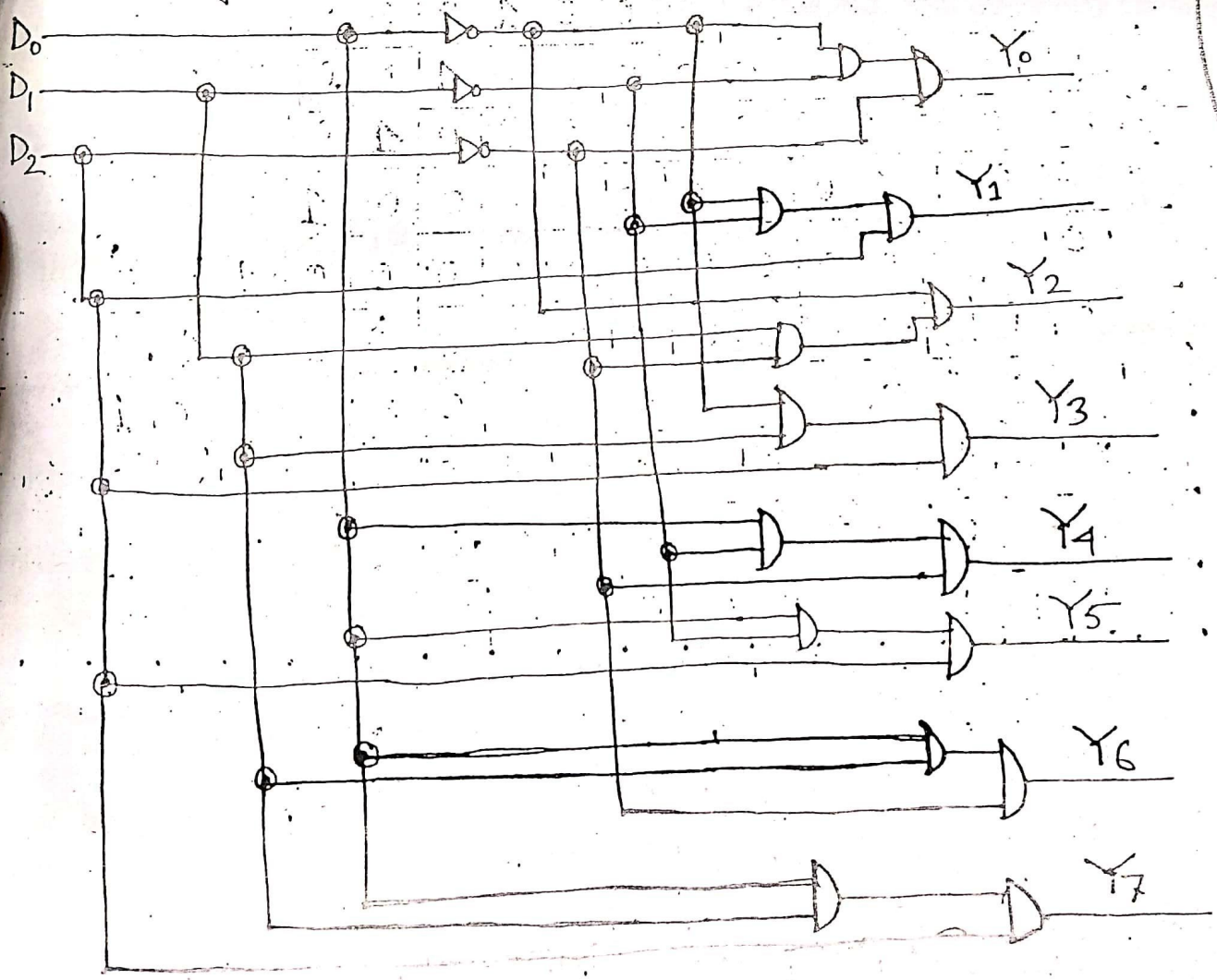
$Y_4 = D_0 \bar{D}_1 \bar{D}_2$

$Y_5 = D_0 \bar{D}_1 D_2$

$Y_6 = D_0 D_1 \bar{D}_2$

$Y_7 = D_0 D_1 D_2$

Circuit Diagram:

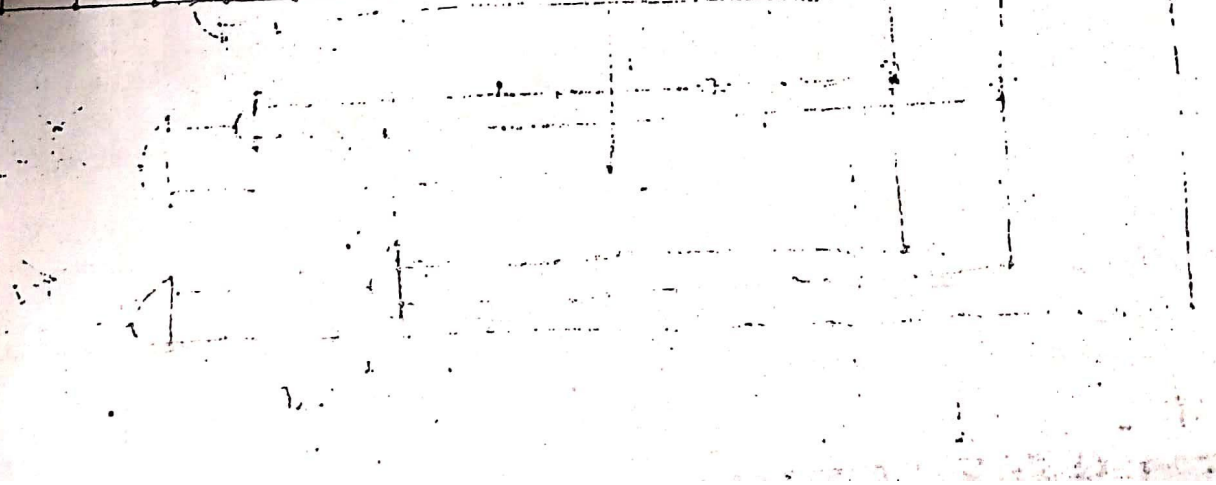


Scanned by CamScanner

4-bit decoder :

Truth table :

D_0	D_1	D_2	D_3	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	Y_8	Y_9	Y_{10}	Y_{11}	Y_{12}	Y_{13}	Y_{14}	Y_{15}
0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1



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$Y_0 = \overline{D_0}\,\overline{D_1}\,\overline{D_2}\,\overline{D_3}$

$Y_1 = \overline{D_0}\,\overline{D_1}\,\overline{D_2}\,D_3$

$Y_2 = \overline{D_0}\,\overline{D_1}\,D_2\,\overline{D_3}$

$Y_3 = \overline{D_0}\,\overline{D_1}\,D_2\,D_3$

$Y_4 = \overline{D_0}\,D_1\,\overline{D_2}\,\overline{D_3}$

$Y_5 = \overline{D_0}\,D_1\,\overline{D_2}\,D_3$

$Y_6 = \overline{D_0}\,D_1\,D_2\,\overline{D_3}$

$Y_7 = \overline{D_0}\,D_1\,D_2\,D_3$

$Y_8 = D_0\,\overline{D_1}\,\overline{D_2}\,\overline{D_3}$

$Y_9 = D_0\,\overline{D_1}\,\overline{D_2}\,D_3$

$Y_{10} = D_0\,\overline{D_1}\,D_2\,\overline{D_3}$

$Y_{11} = D_0\,\overline{D_1}\,D_2\,D_3$

$Y_{12} = D_0\,D_1\,\overline{D_2}\,\overline{D_3}$

$Y_{13} = D_0\,D_1\,\overline{D_2}\,D_3$

$Y_{14} = D_0\,D_1\,D_2\,\overline{D_3}$

$Y_{15} = D_0\,D_1\,D_2\,D_3$

SHUVO
CSE-27

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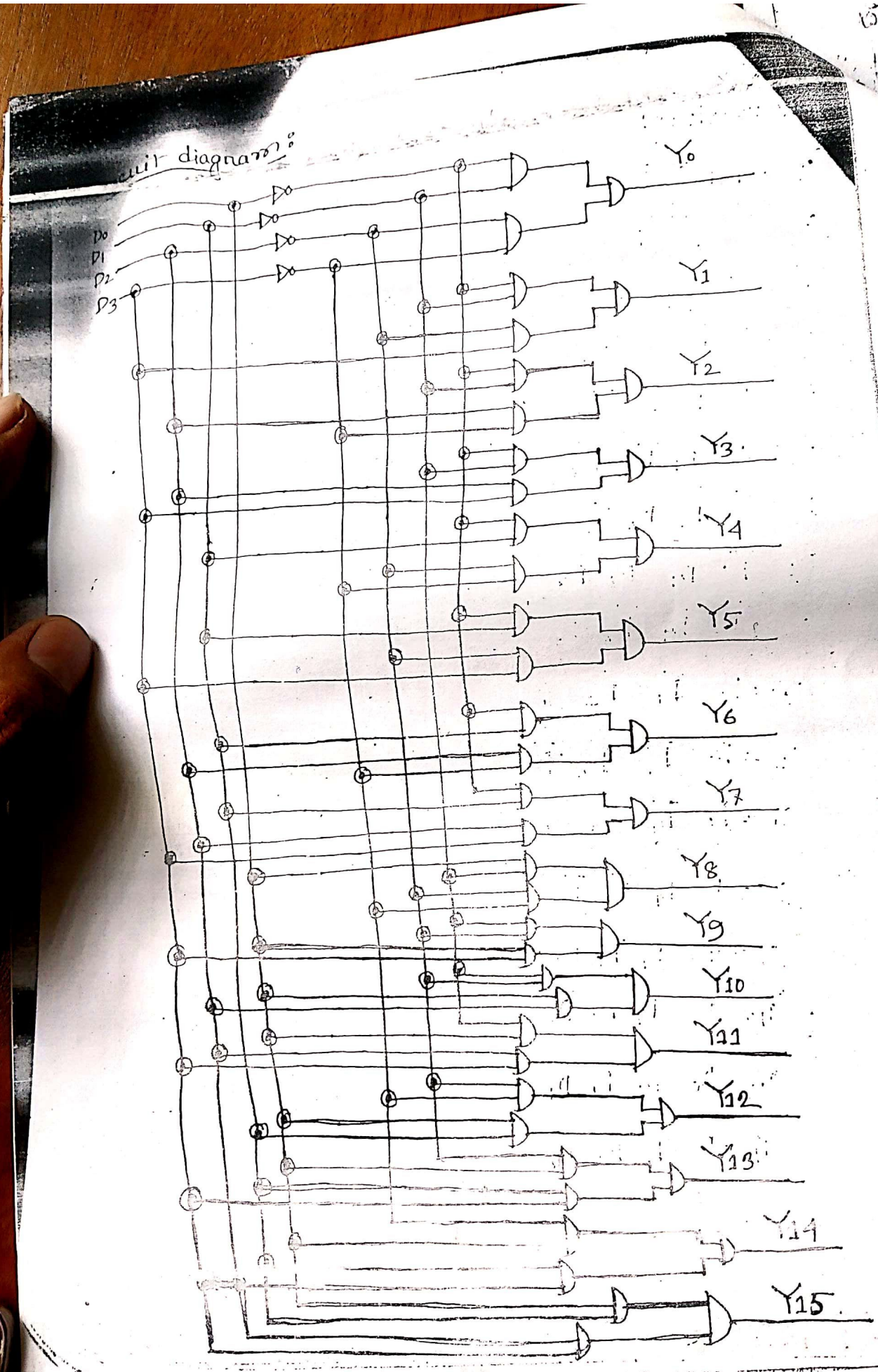
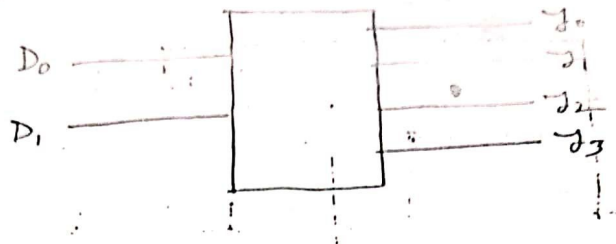


Scanned by CamScanner

2-Bit Decoder:



Truth table:

D_0	D_1	y_0	y_1	y_2	y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Expression:

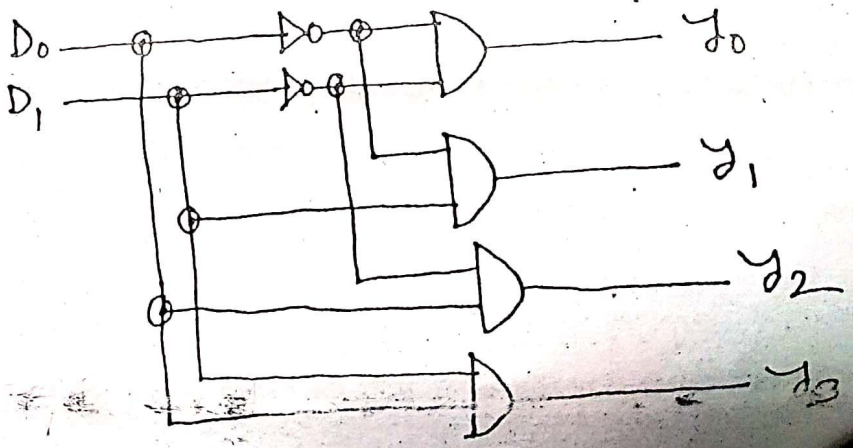
$$y_0 = \overline{D_0}\,\overline{D_1}$$

$$y_1 = \overline{D_0}\,D_1$$

$$y_2 = D_0\,\overline{D_1}$$

$$y_3 = D_0\,D_1$$

Circuit Diagram:



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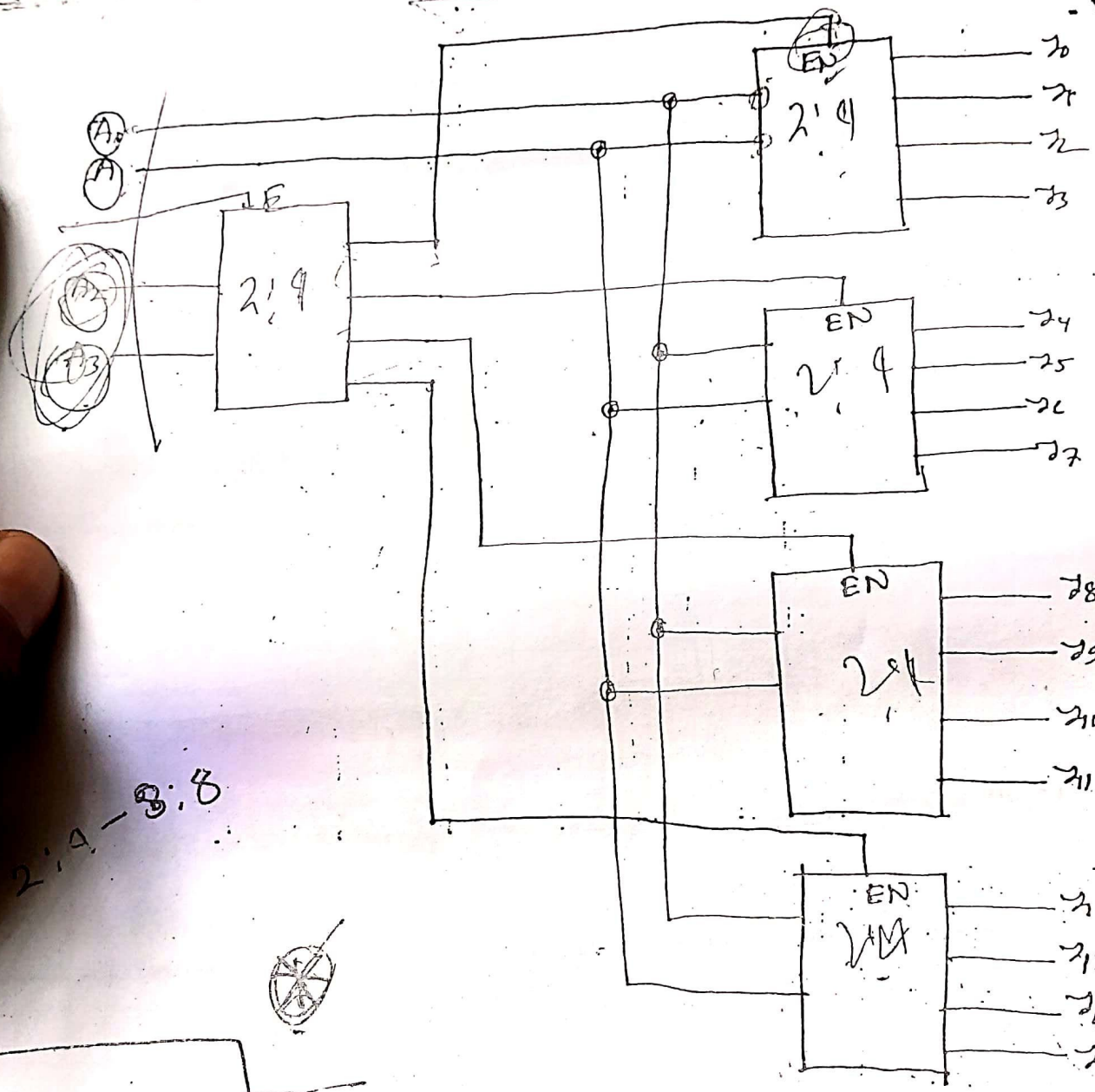
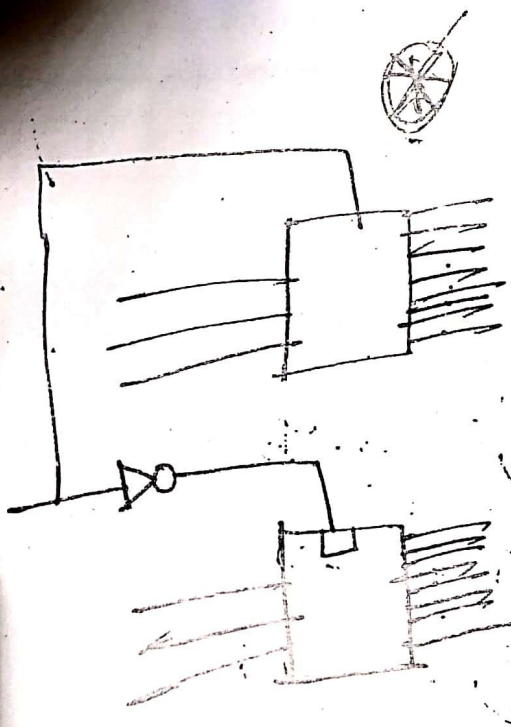
T-2

Q) Determine 4:16 Decoder from 2:4 Dec



2:4 - 3:8



Sapti

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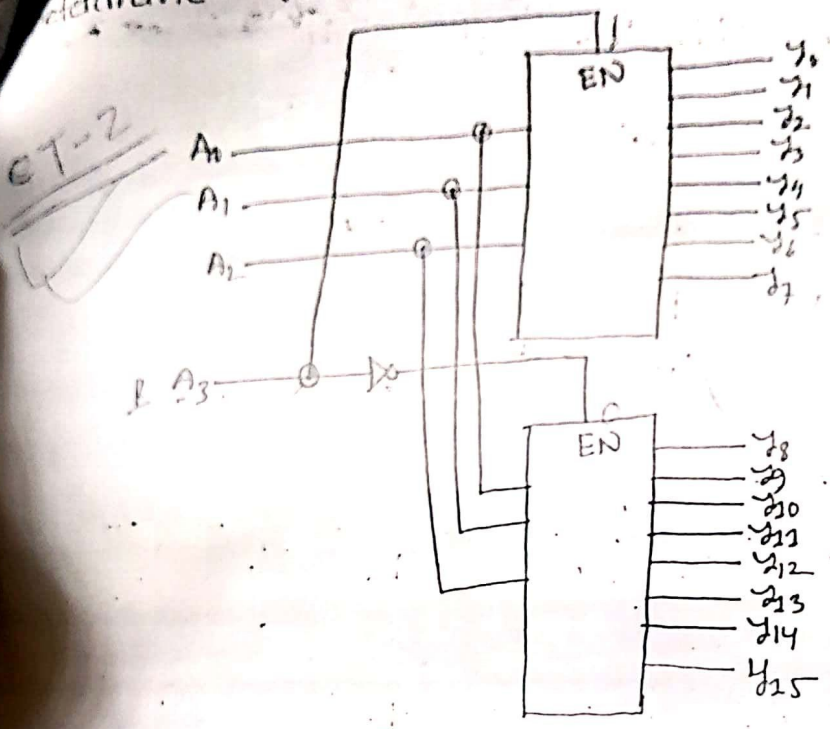
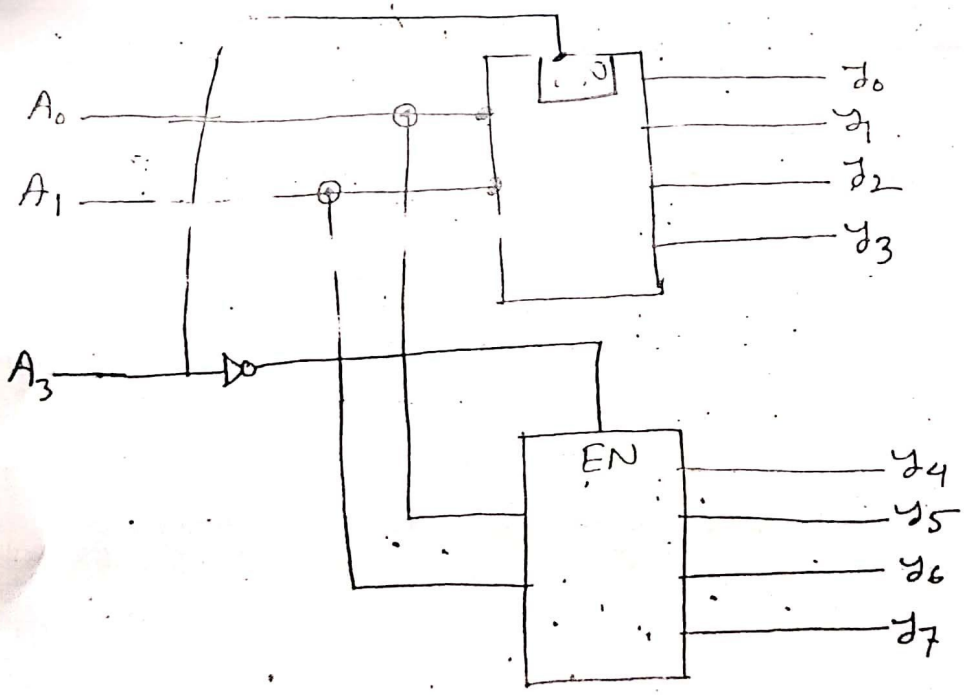
Determine 4:16 Decoder from 3:8 Decoder.



Determine 3:8 Decoder from 2:4 Decoder.



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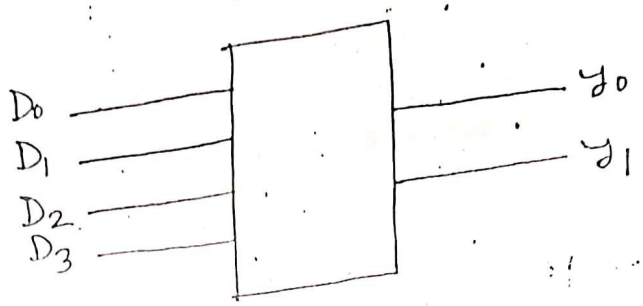
4:2 Encoder

Encoder:



Truth table:

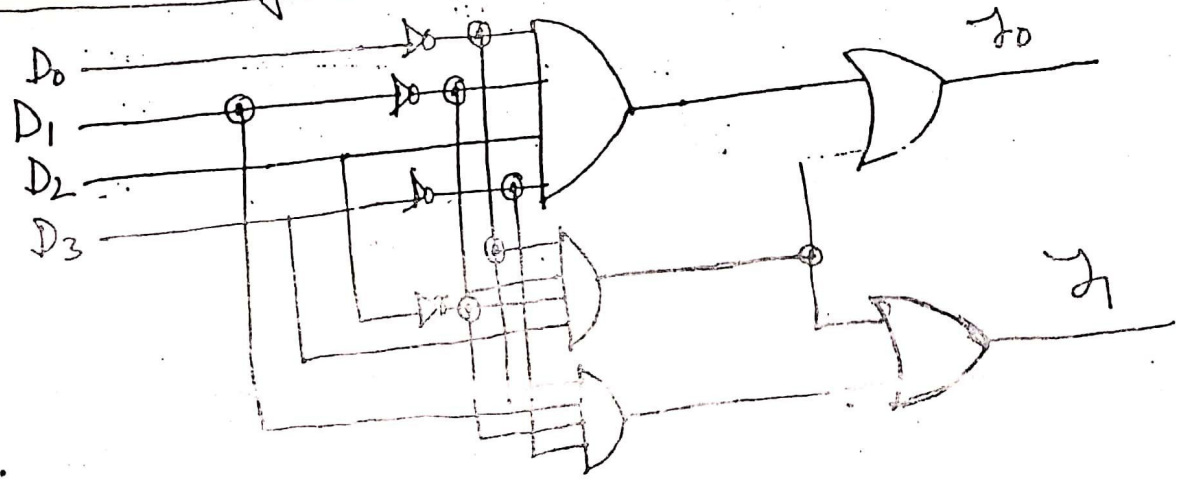
D_0	D_1	D_2	D_3	y_0	y_1
1	0	0	φ	0	0
0	1	φ	φ	0	1
0	0	1	φ	1	0
0	0	0	1	1	1

Expression:

$$y_0 = \overline{D_0}\,\overline{D_1}\,D_2\,\overline{D_3} + \overline{D_0}\,D_1\,\overline{D_2}\,D_3$$

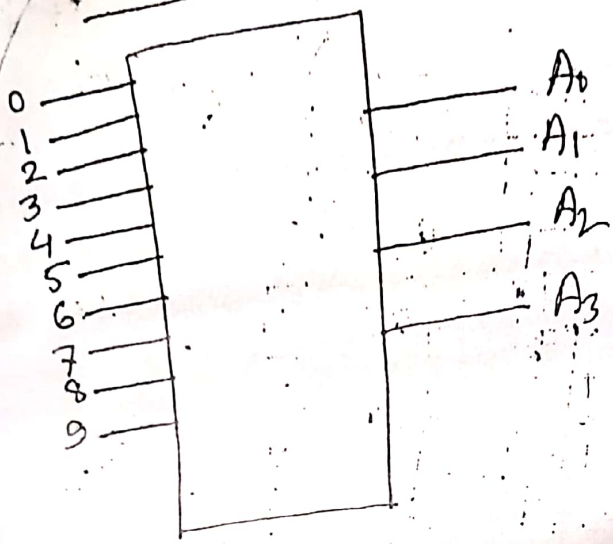
$$y_1 = \overline{D_0}\,D_1\,\overline{D_2}\,\overline{D_3} + \overline{D_0}\,\overline{D_1}\,\overline{D_2}\,D_3$$

Circuit Diagram:



Scanned by CamScanner

10:4 Encoder



Truth table:

input	A_3	A_2	A_1	A_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

Expression:

$A_0 = 1+3+5+7+9$

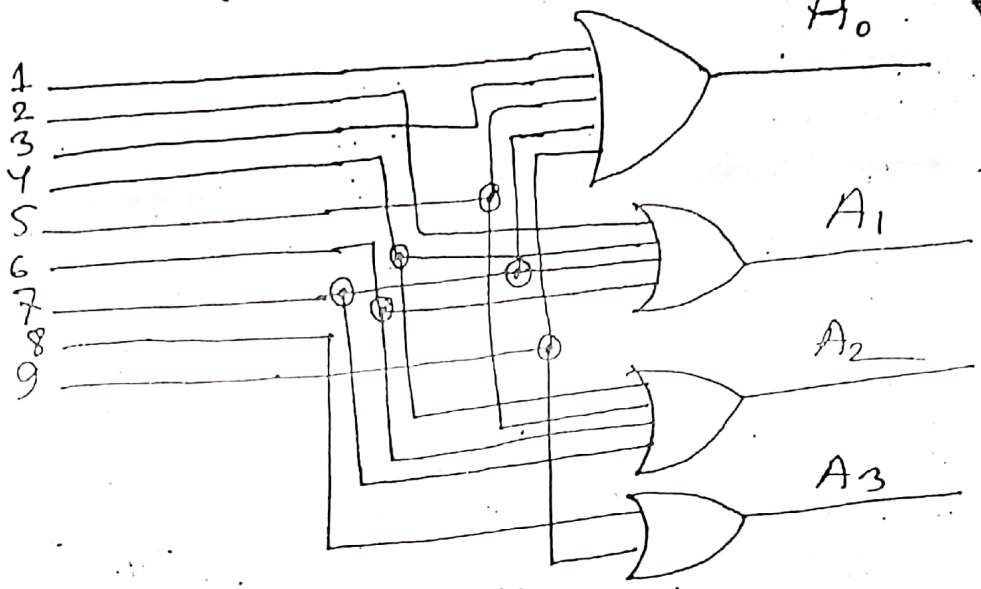
$A_1 = 2+3+6+7$

$A_2 = 4+5+6+7$

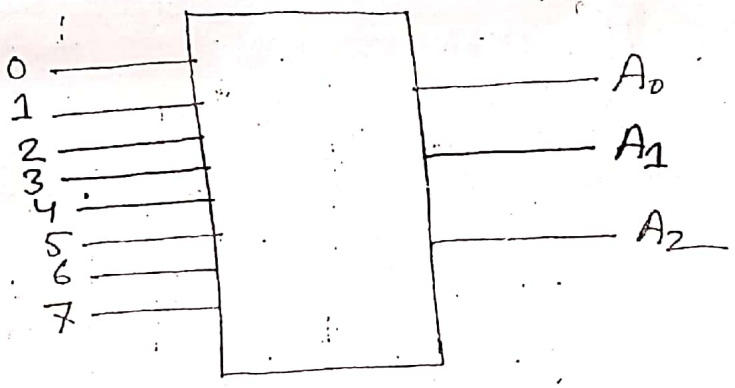
$A_3 = 8+9$

Scanned by CamScanner

Circuit Diagram:



8:3 Encoder



Truth table:

input	A_2	A_1	A_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

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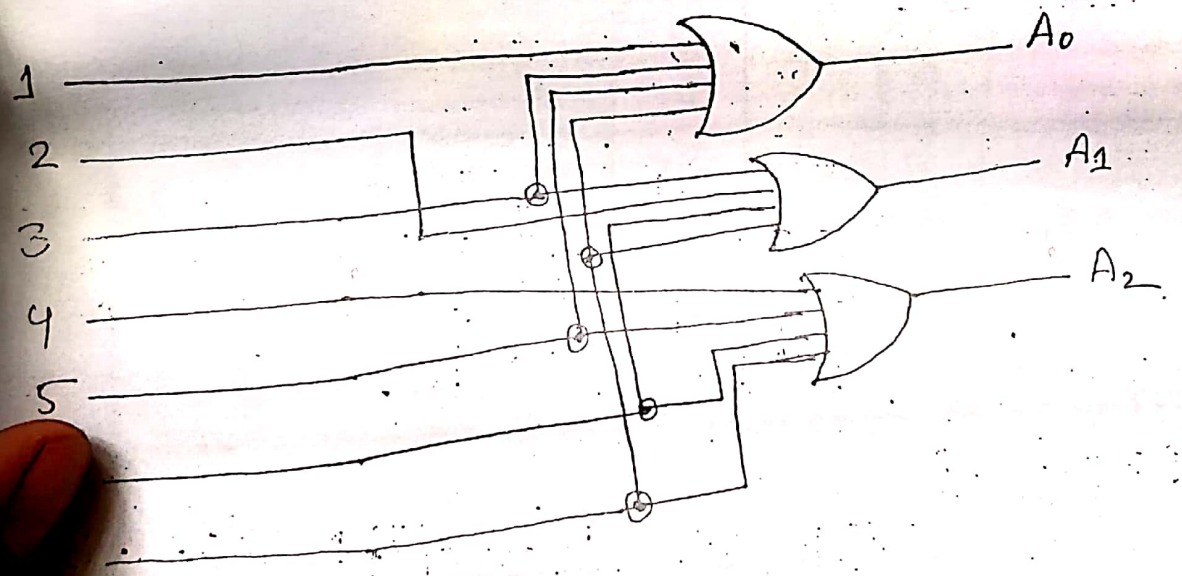
Expression:

$A_0 = 1+3+5+7$.

$A_1 = 2+3+6+7$

$A_2 = 4+5+6+7$.

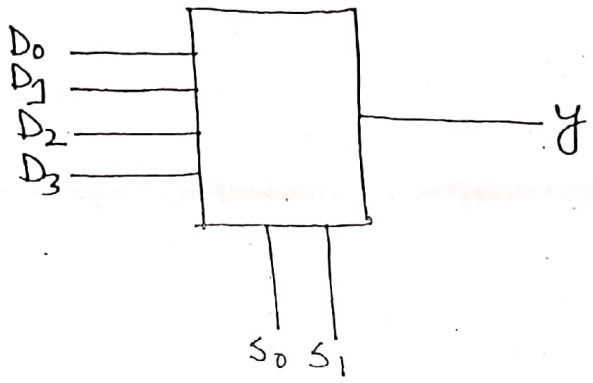
Circuit Diagram:



Scanned by CamScanner

* Multiplexer :

4:1 MUX



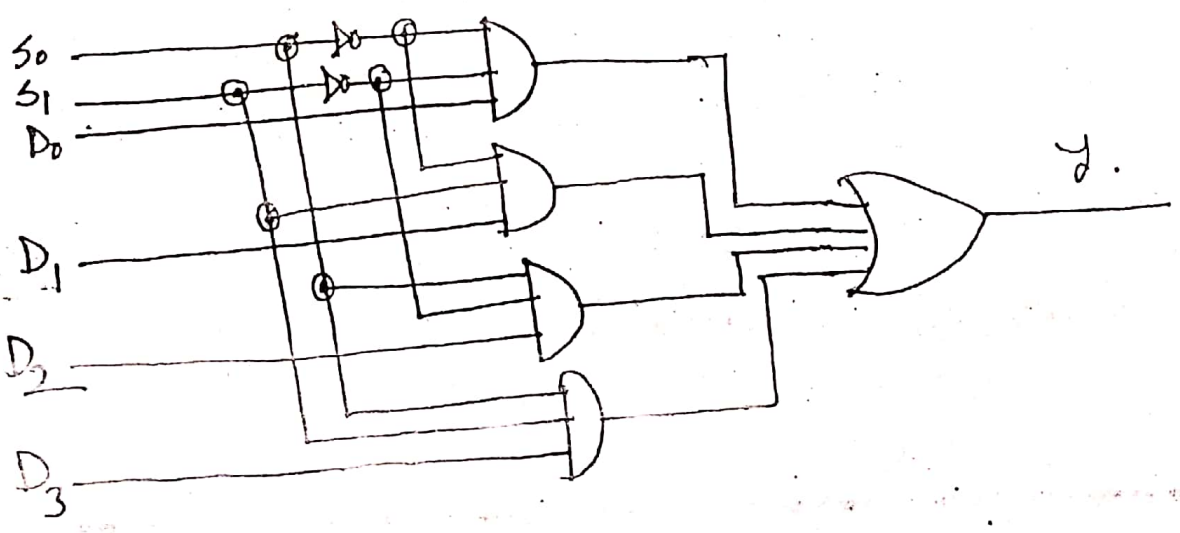
Truth table :

S_0	S_1	y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

Expression :

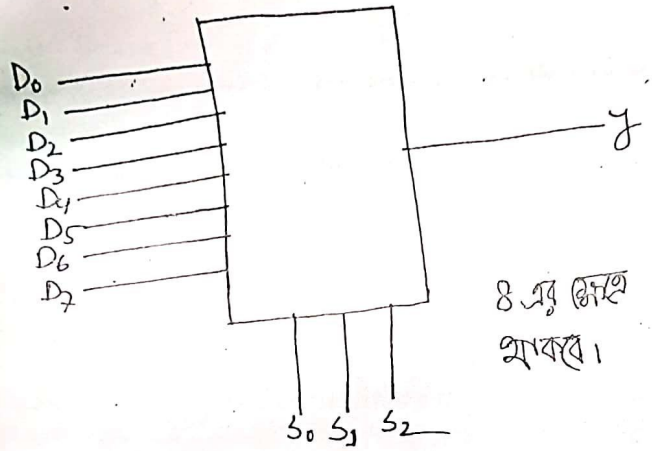
$$y = \overline{S_0}\,\overline{S_1}\,D_0 + \overline{S_0}\,S_1 D_1 + S_0\overline{S_1}\,D_2 + S_0 S_1 D_3$$

Circuit Diagram :



Scanned by CamScanner

8:1 MUX



8 টা ইনপুট 3 টা Selector থাকবে।

Truth table:

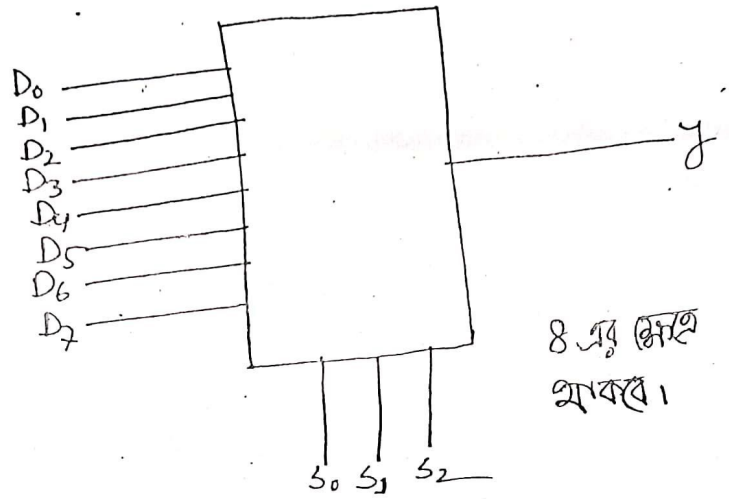
S_2	S_1	S_0	y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

$$\therefore y = \overline{S_2}\,\overline{S_1}\,\overline{S_0} D_0 + \overline{S_2}\,\overline{S_1} S_0 D_1 + \overline{S_2} S_1 \overline{S_0} D_2 + \overline{S_2} S_1 S_2 D_3 + S_2 \overline{S_1}\,\overline{S_0} D_4 + S_2 \overline{S_1} S_0 D_5 + S_2 S_1 \overline{S_0} D_6 + S_2 S_1 S_0 D_7$$

এভাবে সবার Expression গুলো দেখাতে হবে।

Scanned by CamScanner

8:1 MUX



8 টি ইনপুট 3 টি Selector থাকবে।

Truth table:

S_2	S_1	S_0	y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

$$\therefore y = \bar{S_2}\bar{S_1}\bar{S_0}D_0 + \bar{S_2}\bar{S_1}S_0D_1 + \bar{S_2}S_1\bar{S_0}D_2 + \bar{S_2}S_1S_2D_3 + S_2\bar{S_1}\bar{S_0}D_4 + S_2\bar{S_1}S_0D_5 + S_2S_1\bar{S_0}D_6 + S_2S_1S_0D_7$$

এখন এভাবে Expression গুলো লেখতে হবে।

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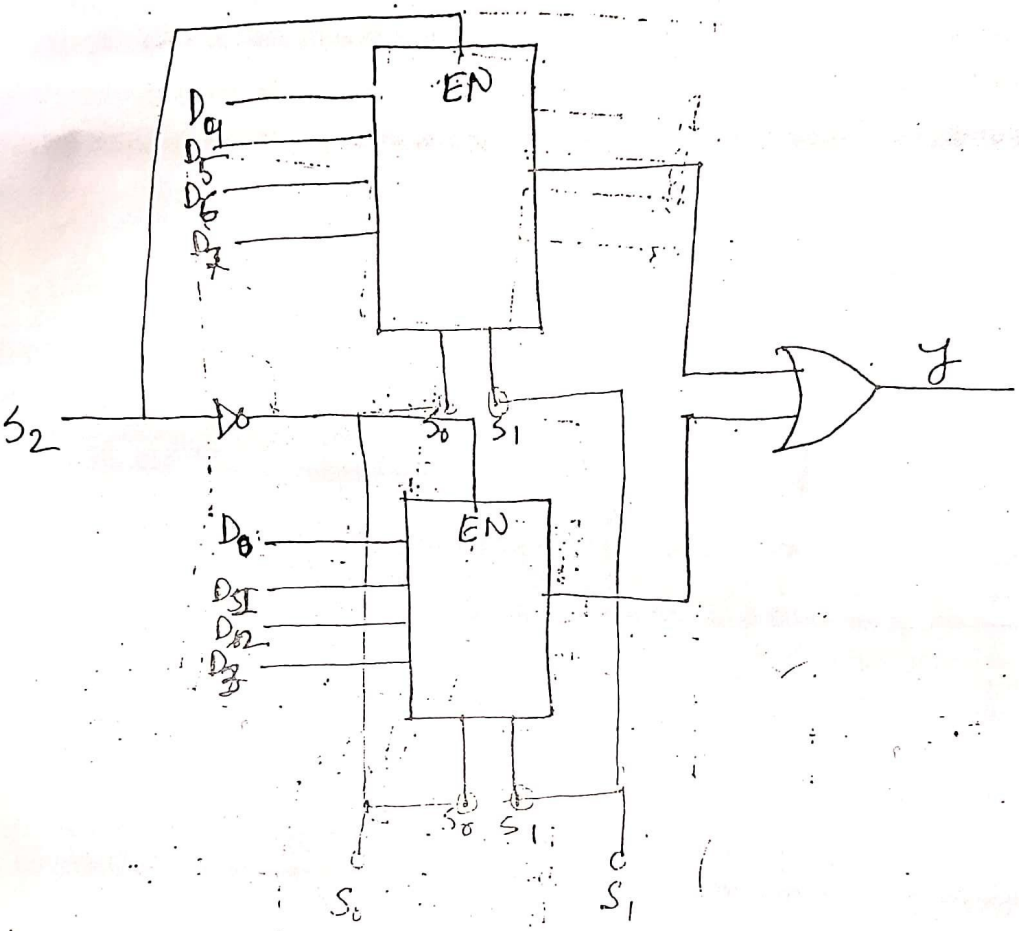
multiplexer from 4:1 Multiplexer



Truth table:

S_2	S_1	S_0	Y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

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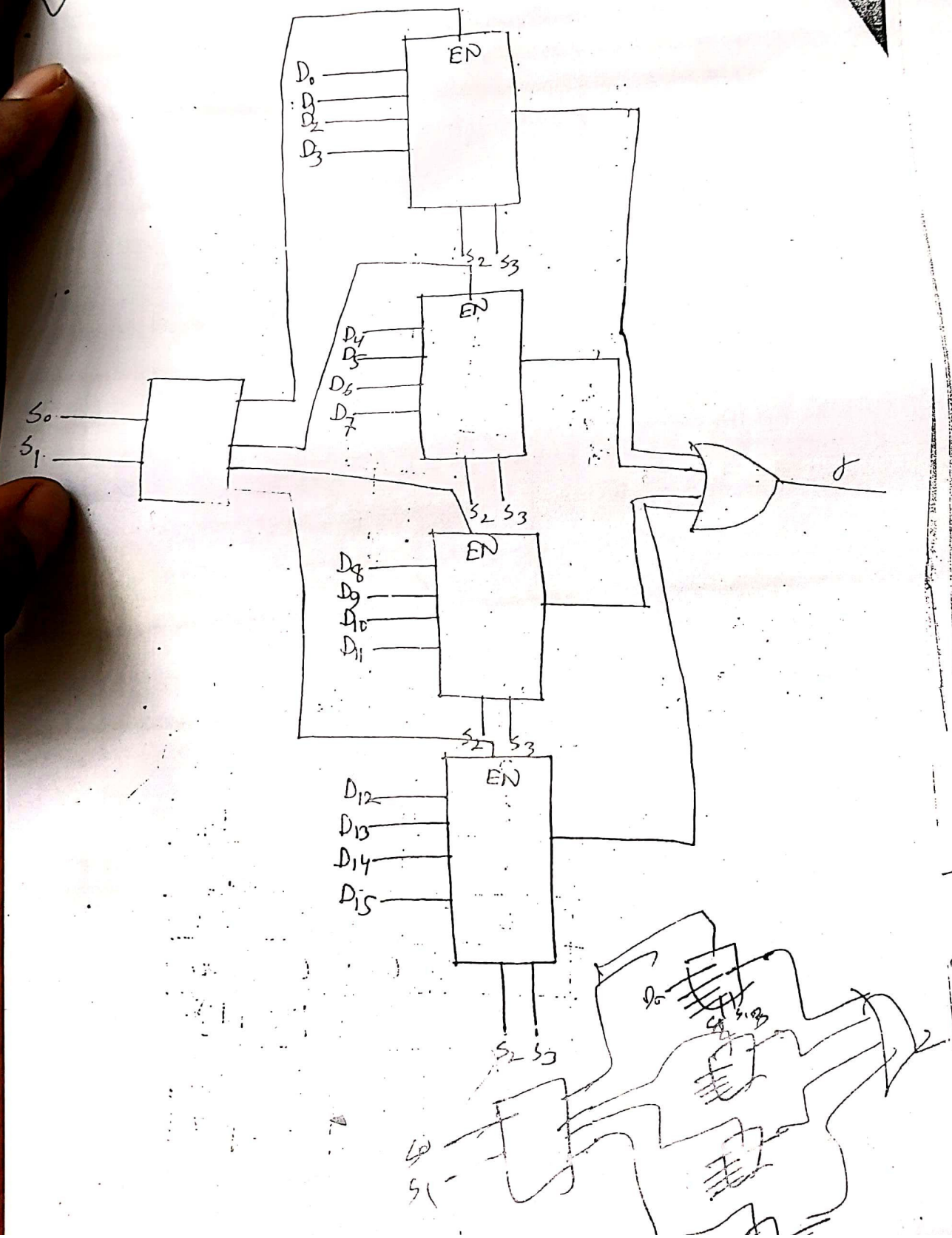
16:1 MUX from 4:1 MUX



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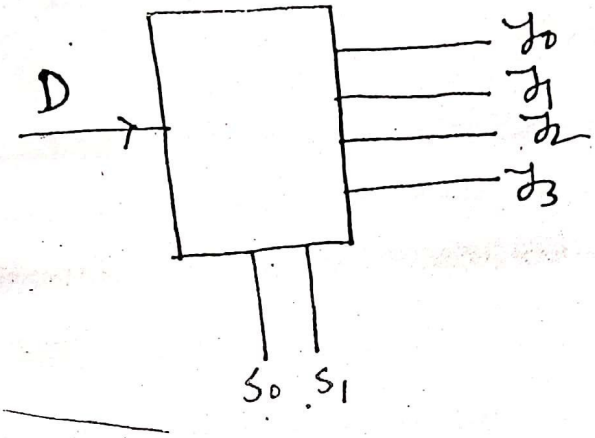
Demultiplexer



Truth table:

S_1	S_0	y_0	y_1	y_2	y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Formula:

$$y_0 = \overline{S_1}\,\overline{S_0}\,D$$

$$y_1 = \overline{S_1} S_0 D$$

$$y_2 = S_1 \overline{S_0} D$$

$$y_3 = S_1 S_0 D$$

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Counters

Counters: when a group of flip-flops are connected together to perform counting operations such a group of flip-flops are called a counter.

Basically counters are of two types based on the way they are clocked.

1. Asynchronous counters.
2. synchronous counter.

The Difference between Asynchronous and synchronous counters are given below:-

Asynchronous	Synchronous
① In this counter all the flip-flops are not triggered simultaneously	① In this counter all the flip-flops are triggered simultaneously.
ey do not have a common clock pulse.	② They have a common clock pulse.
The output of the 1st flip-flop is used as the clock for he 2nd flip-flop.	③ This counters can operate at a much higher input frequencies because of low propagation delay.
This counters are also eferred to as ripple ounters.	④ They have more compier circuitry.

propagation delay.

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mected to
a group of
the way

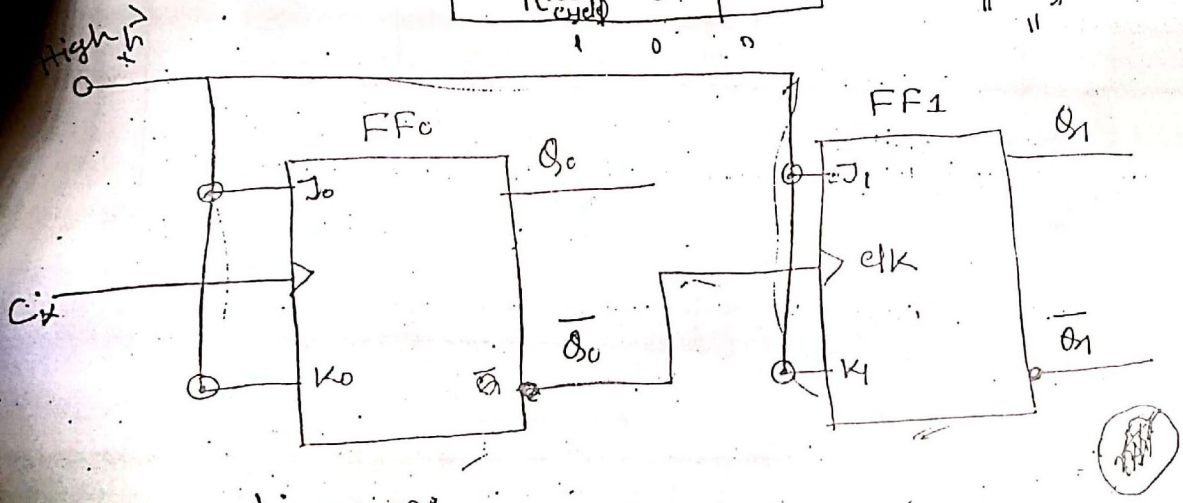
Asynchronous Counters

2- bit asynchronous counters: (MOD-4)

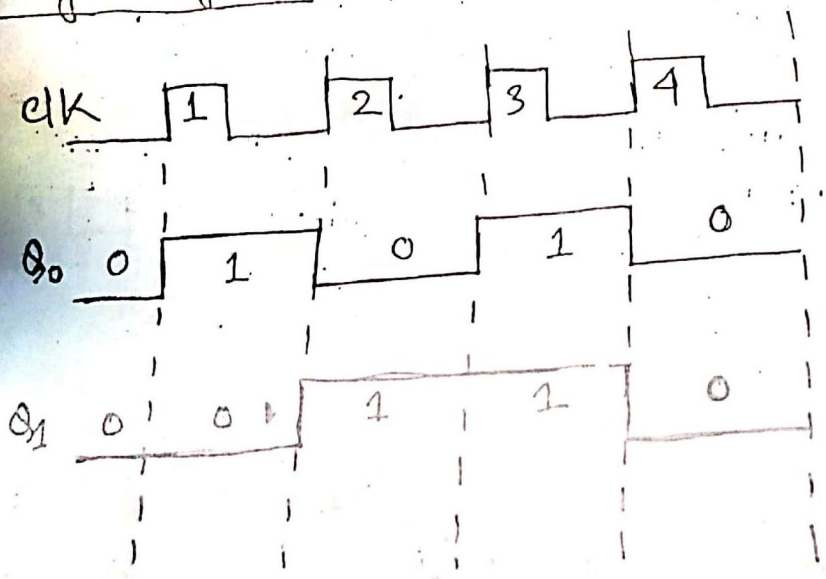
Truth table:

clk	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4 (re cycle)	0	0

$J=K=1$



Timing diagram:



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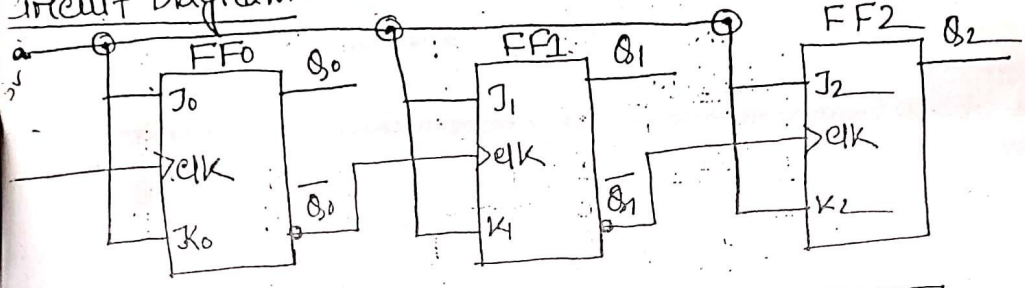
3-bit asynchronous Binary Counter: (MOD-8)

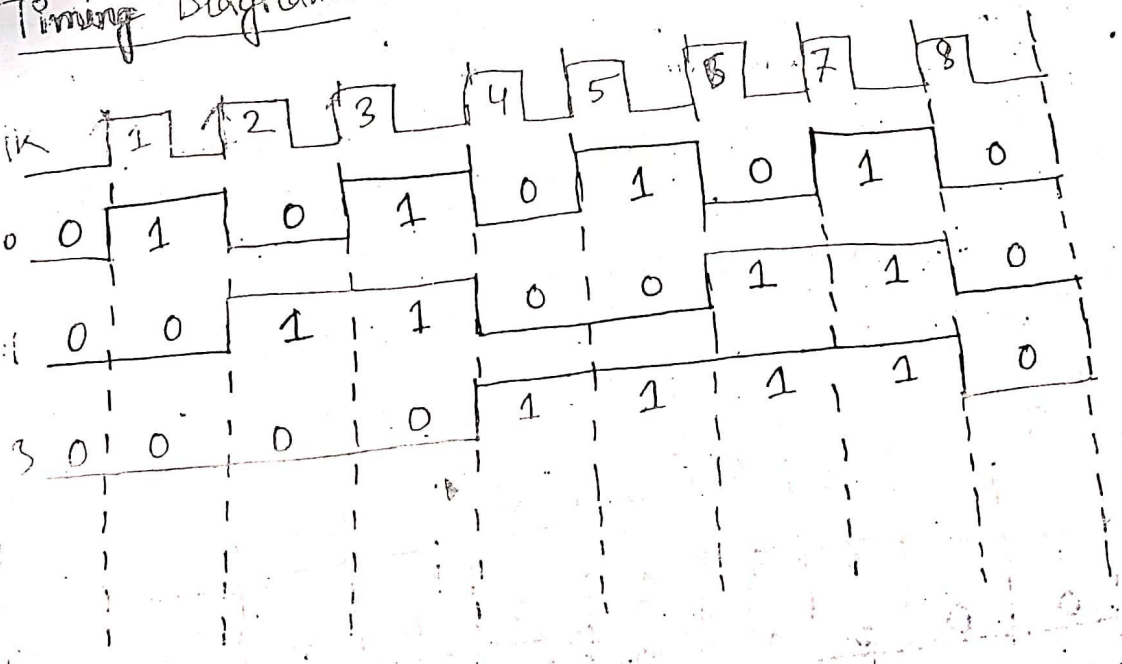
Circuit Diagram:



Truth Table:

clk	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Timing Diagram:



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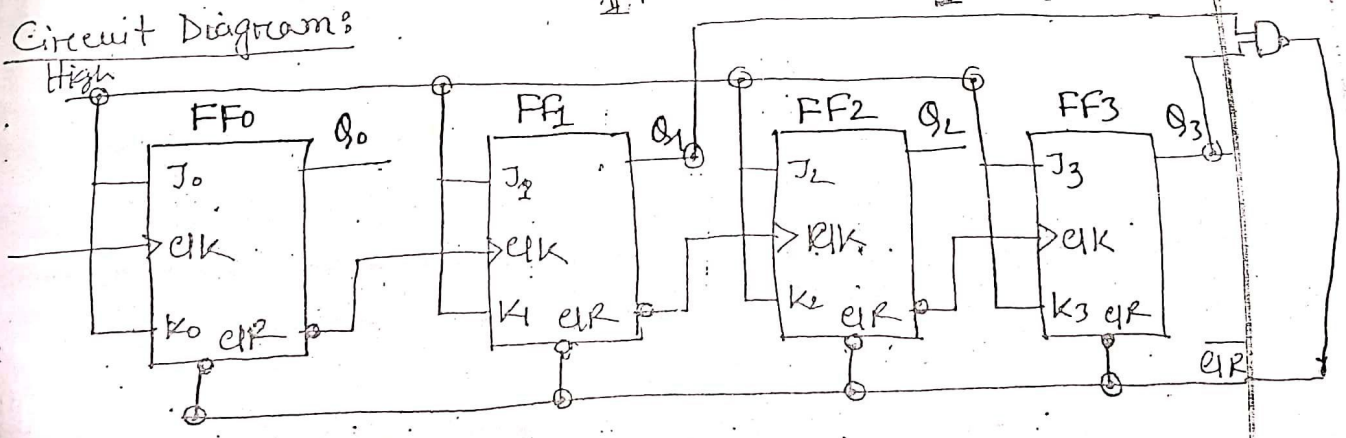
Decade Bed Asynchronous Counter / (Mod-10):

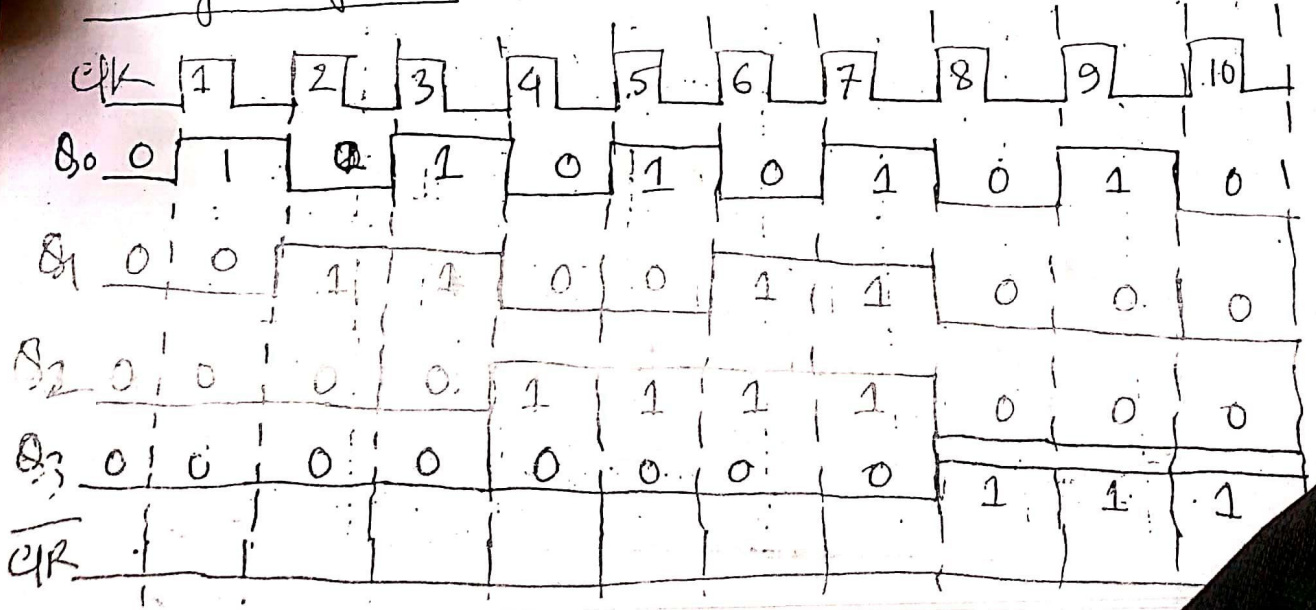
Truth table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0
	1	0	1	0

Circuit Diagram:



Timing Diagram:



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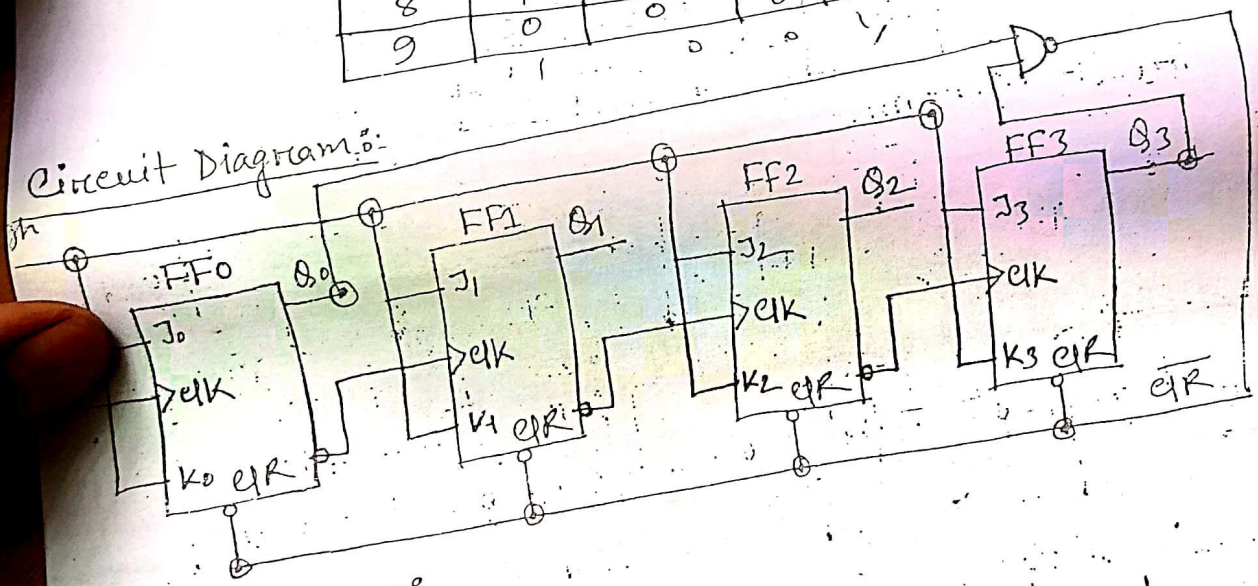
MOD-9 Asynchronous counter:

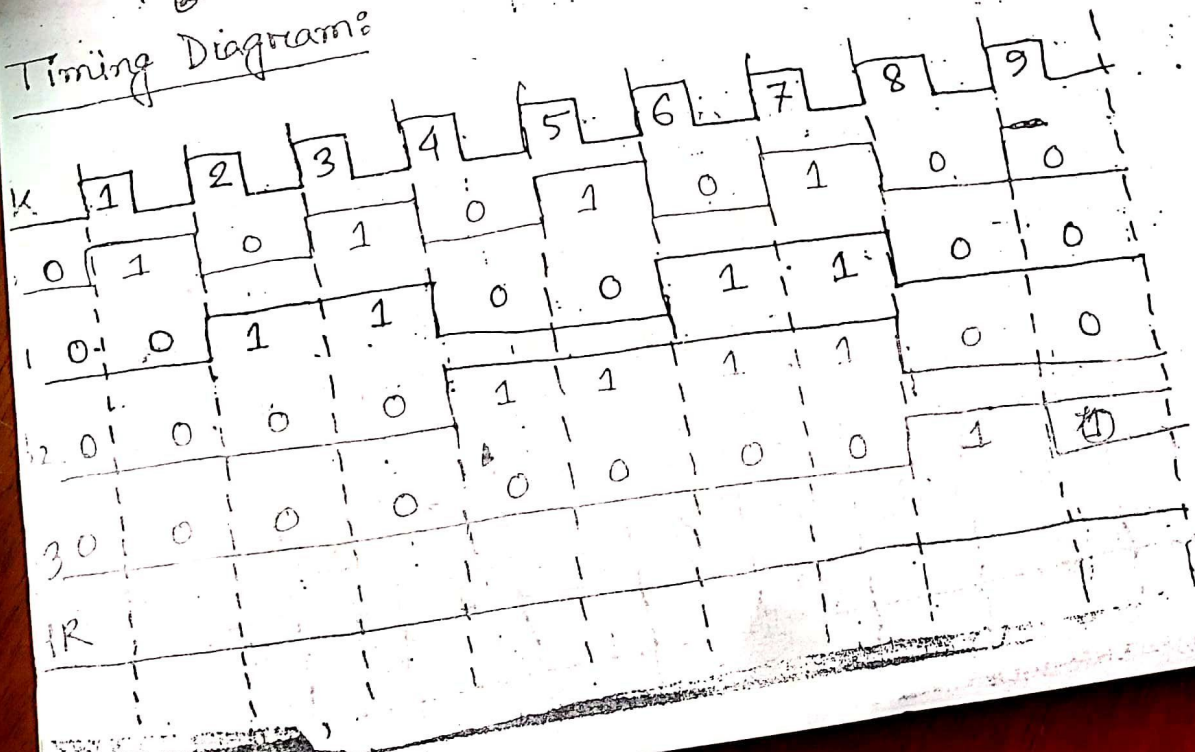
Truth table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	0	0	0	1

Circuit Diagram:



Timing Diagram:

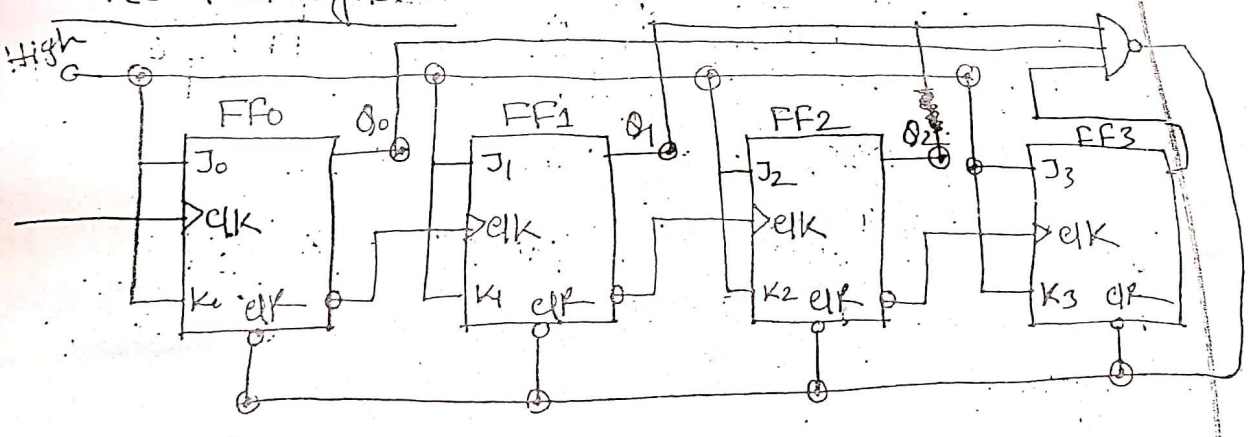

clk
1
2
3
4
5
6
7
8
9
0
1
0
1
0
1
0
1
0
0
1
0
1
0
1
0
0
1
1
0
0
1
1
0
0
0
0
1
1
1
1
0
0
0
0
0
0
0
0
1
1
0
0
0
0
0
0
0
0

* MOD-11 Asynchronous Counter:

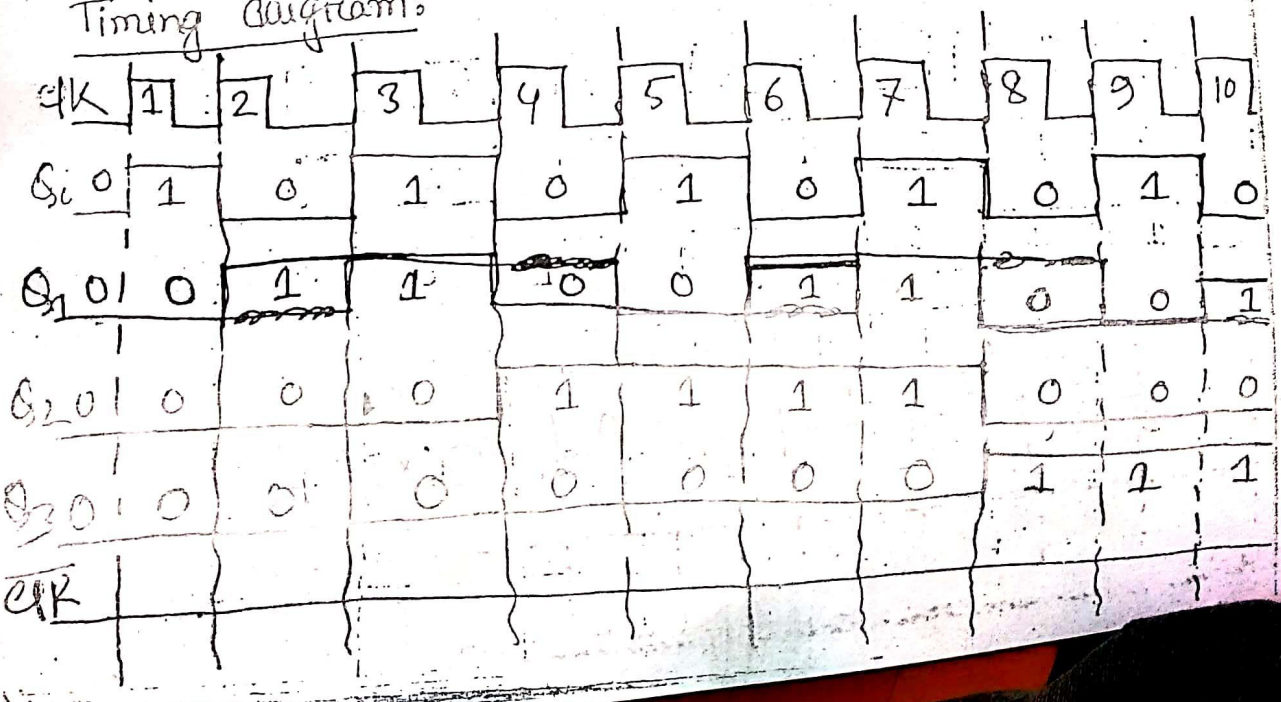
Truth table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	0	0	0	0

Circuit Diagram:



Timing daigram:



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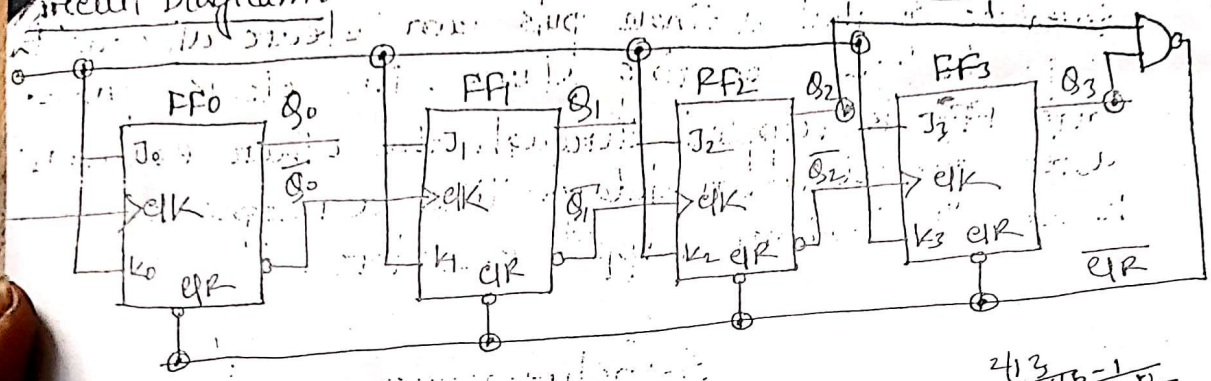
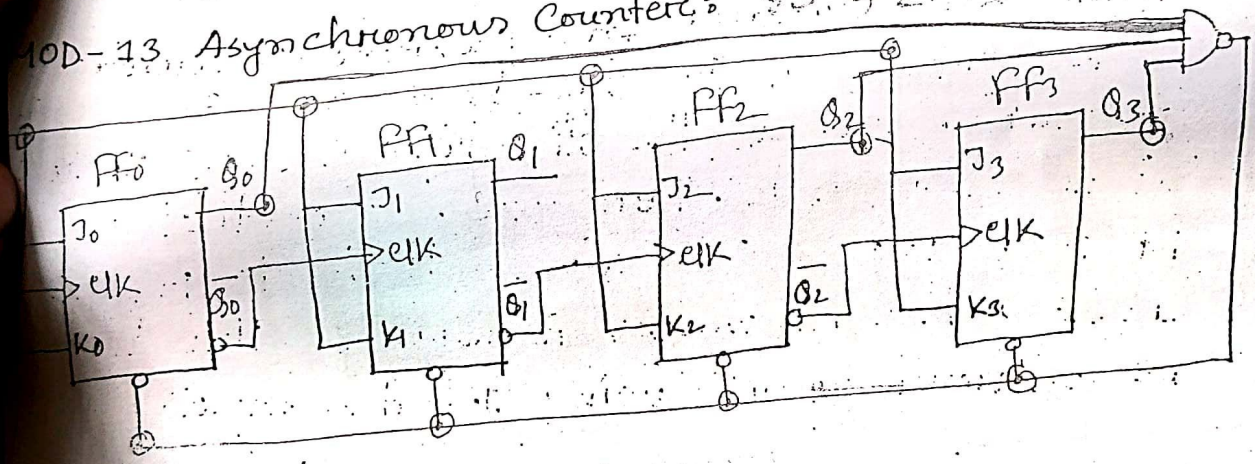
12 Asynchronous Counter:

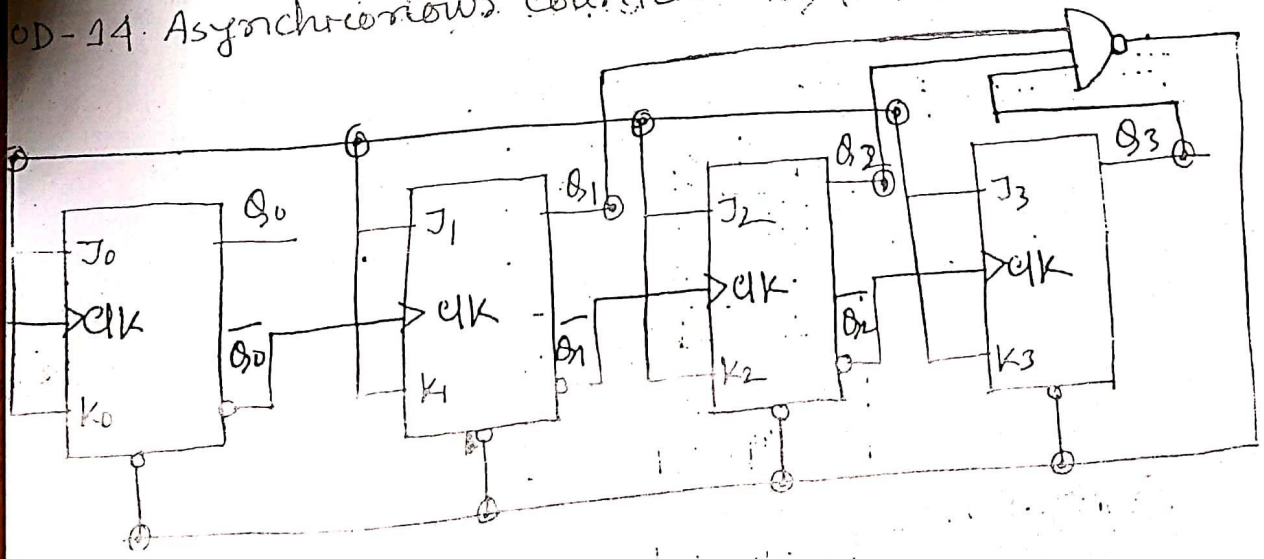
Circuit Diagram:



13. Asynchronous Counter:



14. Asynchronous counter:



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Ripple Counter: Usually all the clear inputs are co... together so that a single puls can clear all the ... flops before counting starts. The clock pulse fe... into FF0 is rippled through the other counters ... after propagation delays like a ripple on wate... hence the name ripple counter.

Synchronous Counter

Synchronous counter: In synchronous counter the clock inputs of all the flip-flops are connected togather. Thus all the flip-flops change their sta... at the same time. Therefore, to control when a FF is to toggle and when it is to remain unchanged, some means is required. This is accompli... by additional circuit using J and K inputs.

2-Bit Synchronous Counter: (MOD-4)

Truth Table:

Clk	Q_1	Q_0
Initially	0	0
1	0	1
2	1	0
3	1	1
4th cycle	0	0

Expression: $J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

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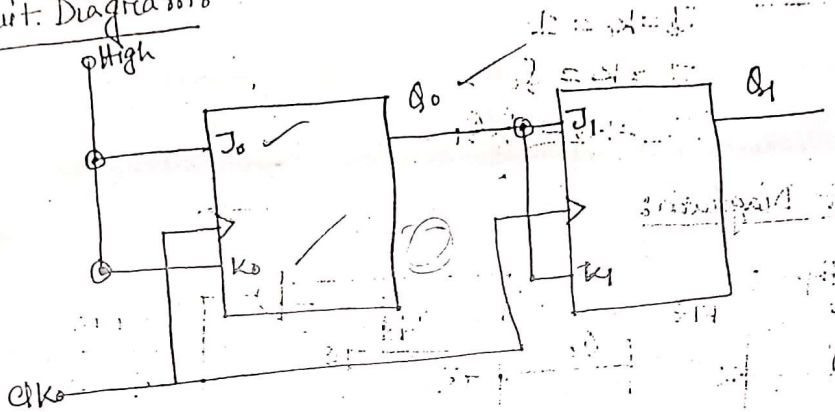
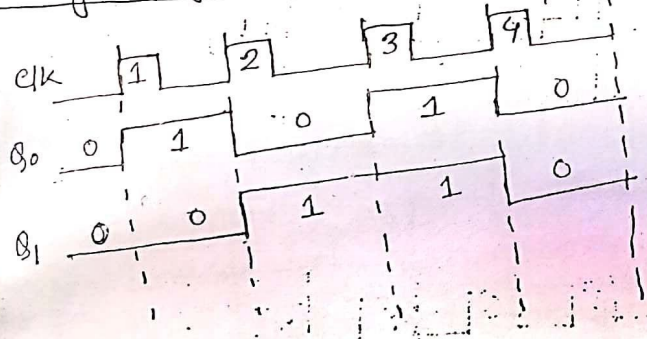
Circuit Diagram:



Timing Diagram:



3-bit synchronous Counter: (MOD-8)

Truth Table:

Clk	Q_2	Q_1	Q_0
Initially	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
0	0	0	0

$J_0 = K_0 = Q_1 = 1$

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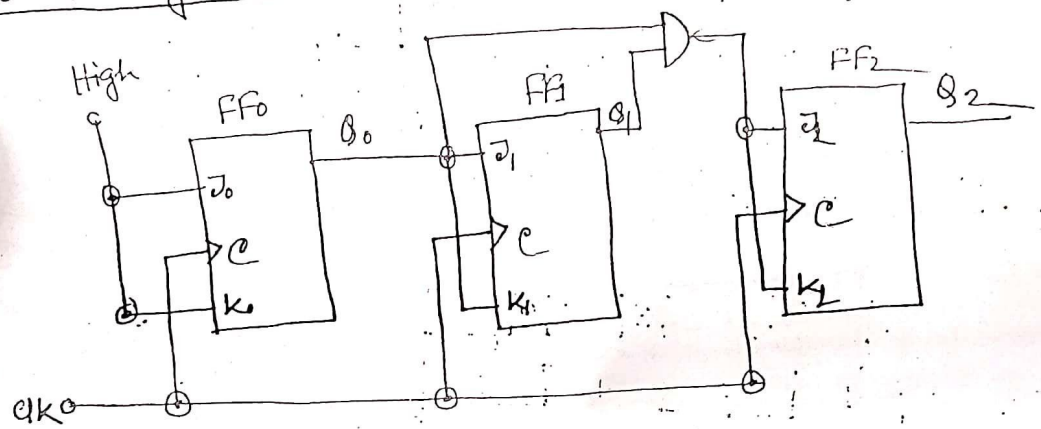
Expression:

$J_0 = K_0 = 1$

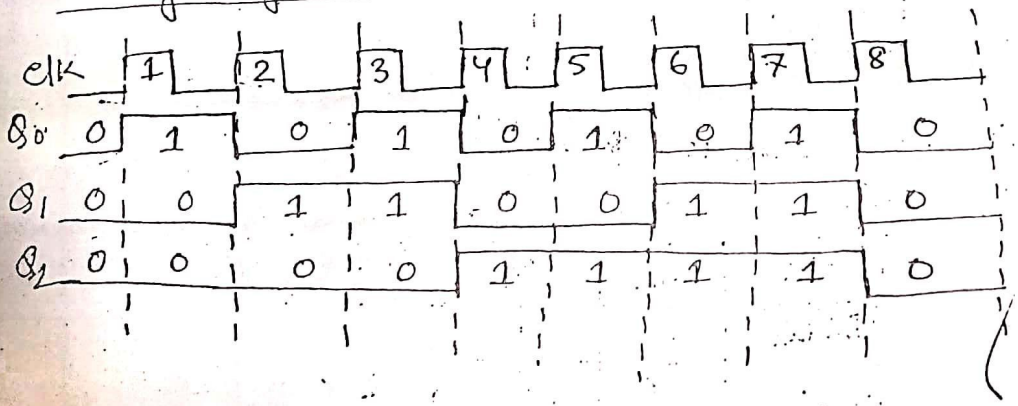
$J_1 = K_1 = Q_0$

$J_2 = K_2 = Q_0 Q_1$

Circuit Diagram:



Timing Diagram:



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4 bit Synchronous Counter : (MOD-16)

Truth Table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1
16	0	0	0	0

Expression:

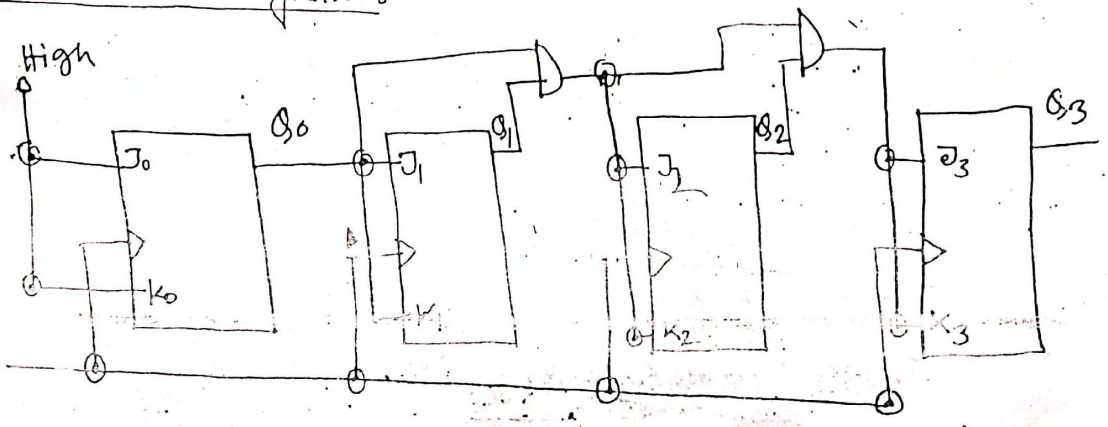
$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

$J_2 = K_2 = Q_0 Q_1$

$J_3 = K_3 = Q_0 Q_1 Q_2$

Circuit Diagram:



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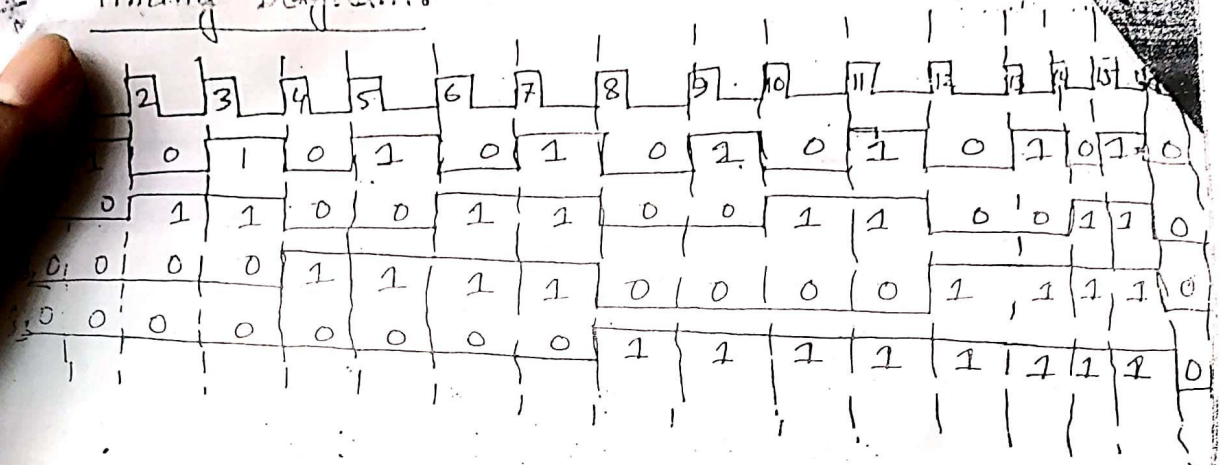
Timing Diagram:



4-bit synchronous Decade Counter: (MOD-10)

Truth Table:

clk	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0

Expression:

$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0 Q_3$

$J_2 = K_2 = Q_0 Q_1$

$J_3 = K_3 = Q_0 Q_1 Q_2 + Q_0 Q_3$

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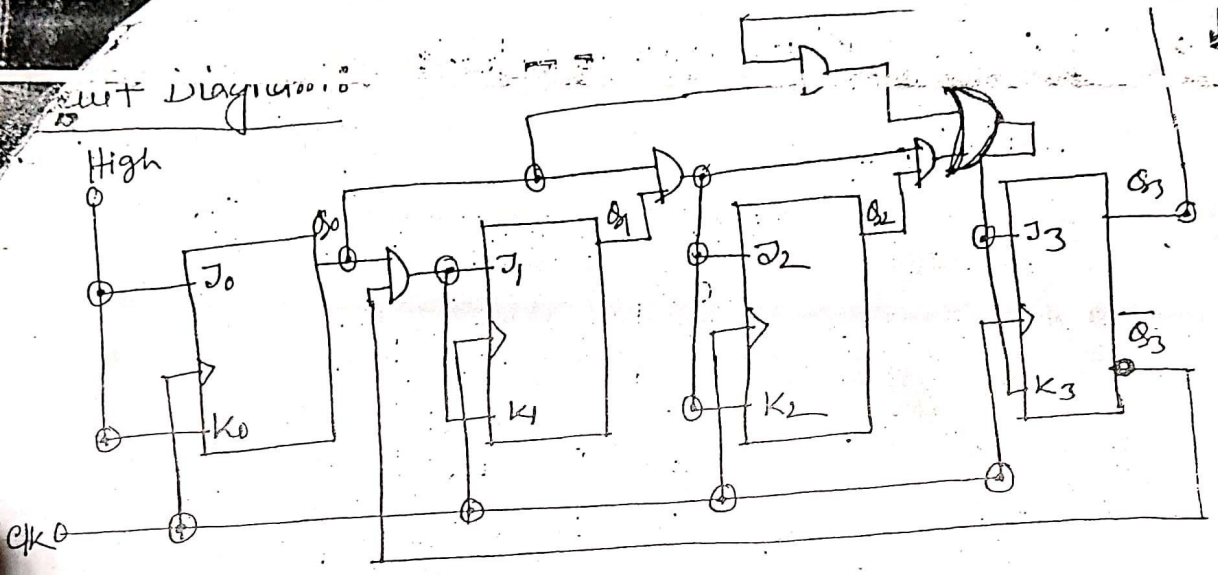
Circuit Diagram:



Timing Diagram:

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